

TABLES

Table I Names, symbols and atomic numbers of the elements (see also Section IR-3.1)

<i>Name</i>	<i>Symbol</i>	<i>Atomic number</i>	<i>Name</i>	<i>Symbol</i>	<i>Atomic number</i>
actinium	Ac	89	indium	In	49
aluminium ^a	Al	13	iodine	I	53
americium	Am	95	iridium	Ir	77
antimony	Sb ^b	51	iron	Fe ^g	26
argon	Ar	18	krypton	Kr	36
arsenic	As	33	lanthanum	La	57
astatine	At	85	lawrencium	Lr	103
barium	Ba	56	lead	Pb ^h	82
berkelium	Bk	97	lithium	Li	3
beryllium	Be	4	lutetium	Lu	71
bismuth	Bi	83	magnesium	Mg	12
bohrium	Bh	107	manganese	Mn	25
boron	B	5	meitnerium	Mt	109
bromine	Br	35	mendelevium	Md	101
cadmium	Cd	48	mercury	Hg ⁱ	80
caesium ^c	Cs	55	molybdenum	Mo	42
calcium	Ca	20	neodymium	Nd	60
californium	Cf	98	neon	Ne	10
carbon	C	6	neptunium	Np	93
cerium	Ce	58	nickel	Ni	28
chlorine	Cl	17	niobium	Nb	41
chromium	Cr	24	nitrogen ^j	N	7
cobalt	Co	27	nobelium	No	102
copper	Cu ^d	29	osmium	Os	76
curium	Cm	96	oxygen	O	8
darmstadtium	Ds	110	palladium	Pd	46
dubnium	Db	105	phosphorus	P	15
dysprosium	Dy	66	platinum	Pt	78
einsteinium	Es	99	plutonium	Pu	94
erbium	Er	68	polonium	Po	84
europium	Eu	63	potassium	K ^k	19
fermium	Fm	100	praseodymium	Pr	59
fluorine	F	9	promethium	Pm	61
francium	Fr	87	protactinium	Pa	91
gadolinium	Gd	64	radium	Ra	88
gallium	Ga	31	radon	Rn	86
germanium	Ge	32	rhenium	Re	75
gold	Au ^e	79	rhodium	Rh	45
hafnium	Hf	72	roentgenium	Rg	111
hassium	Hs	108	rubidium	Rb	37
helium	He	2	ruthenium	Ru	44
holmium	Ho	67	rutherfordium	Rf	104
hydrogen	H ^f	1	samarium	Sm	62

Table I *Continued*

<i>Name</i>	<i>Symbol</i>	<i>Atomic number</i>	<i>Name</i>	<i>Symbol</i>	<i>Atomic number</i>
scandium	Sc	21	thorium	Th	90
seaborgium	Sg	106	thulium	Tm	69
selenium	Se	34	tin	Sn ^o	50
silicon	Si	14	titanium	Ti	22
silver	Ag ^l	47	tungsten	W ^p	74
sodium	Na ^m	11	uranium	U	92
strontium	Sr	38	vanadium	V	23
sulfur ⁿ	S	16	xenon	Xe	54
tantalum	Ta	73	ytterbium	Yb	70
technetium	Tc	43	yttrium	Y	39
tellurium	Te	52	zinc	Zn	30
terbium	Tb	65	zirconium	Zr	40
thallium	Tl	81			

^a The alternative spelling ‘aluminum’ is commonly used.

^b The element symbol Sb derives from the name stibium.

^c The alternative spelling ‘cesium’ is commonly used.

^d The element symbol Cu derives from the name cuprum.

^e The element symbol Au derives from the name aurum.

^f The hydrogen isotopes ²H and ³H are named deuterium and tritium, respectively, for which the symbols D and T may be used. However, ²H and ³H are preferred (see Section IR-3.3.2).

^g The element symbol Fe derives from the name ferrum.

^h The element symbol Pb derives from the name plumbum.

ⁱ The element symbol Hg derives from the name hydrargyrum.

^j The name azote provides the root ‘az’ for nitrogen.

^k The element symbol derives K from the name kalium.

^l The element symbol Ag derives from the name argentum.

^m The element symbol Na derives from the name natrium.

ⁿ The name theion provides the root ‘thi’ for sulfur.

^o The element symbol Sn derives from the name stannum.

^p The element symbol W derives from the name wolfram.

Table II *Temporary names and symbols for elements of atomic number greater than 111^a*

<i>Atomic number</i>	<i>Name^b</i>	<i>Symbol</i>
112	ununbium	Uub
113	ununtrium	Uut
114	ununquadium	Uuq
115	ununpentium	Uup
116	ununhexium	Uuh
117	ununseptium	Uus
118	ununoctium	Uuo
119	ununennium	Uue
120	unbinilium	Ubn
121	unbiunium	Ubu
130	untrinilium	Utn
140	unquadnilium	Uqn
150	unpentnilium	Upn
160	unhexnilium	Uhn
170	unseptnilium	Usn
180	unoctnilium	Uon
190	unennilium	Uen
200	binilnilium	Bnn
201	binilunium	Bnu
202	binilbium	Bnb
300	trinilnilium	Tnn
400	quadnilnilium	Qnn
500	pentnilnilium	Pnn
900	ennilnilium	Enn

^a These names are used only when the permanent name has not yet been assigned by IUPAC (see Section IR-3.1.1).
^b One may also write, for example, ‘element 112’.

Table III *Suffixes and endings*^a

a	Terminal vowel of prefixes indicating replacement of: carbon atoms by atoms of other elements in skeletal replacement nomenclature (Section IR-6.2.4.1) and Hantzsch–Widman nomenclature (Section IR-6.2.4.3), <i>e.g.</i> ‘oxa’, ‘aza’; boron atoms by atoms of other elements in boron hydride-based nomenclature (Section IR-6.2.4.4), <i>e.g.</i> ‘carba’, ‘thia’; heteroatoms by carbon atoms in natural product nomenclature (prefix ‘carba’). See Table X for ‘a’ prefixes for all elements.
ane	Ending of names of neutral saturated parent hydrides of elements of Groups 13–17, <i>e.g.</i> thallane, cubane, cyclohexane, cyclohexasilane, diphosphane, tellane, λ^4-tellane. <i>Cf.</i> Section IR-6.2.2 and Table IR-6.1. Last part of endings of a number of parent names of saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>i.e.</i> of ‘irane’, ‘etane’, ‘olane’, ‘ane’, ‘inane’, ‘epane’, ‘ocane’, ‘onane’ and ‘ecane’ (see Section IR-6.2.4.3).
anide	Combined ending of names of anions resulting from the removal of a hydron from a parent hydride with an ‘ane’ name, formed by adding the suffix ‘ide’, <i>e.g.</i> methanide, CH_3^- . <i>Cf.</i> Section IR-6.4.4.
anium	Combined ending of names of cations resulting from the addition of a hydron to a parent structure with an ‘ane’ name, formed by adding the suffix ‘ium’, <i>e.g.</i> phosphanium, PH_4^+ . <i>Cf.</i> Section IR-6.4.1.
ano	Ending resulting from the change of the ‘ane’ ending in names of parent hydrides to form prefixes denoting bridging divalent substituent groups, <i>e.g.</i> diazano, $-\text{HNNH}-$.
ate	General ending of additive names of anions, <i>e.g.</i> tetrahydridoaluminate(1–), $[\text{AlH}_4]^-$. <i>Cf.</i> Section IR-7.1.4 and Table X. Ending of names of anions and esters of inorganic oxoacids having the ‘ic’ ending in the acid name, <i>e.g.</i> nitrate, phosphonate, trimethyl phosphate, and of anions and esters of organic acids, <i>e.g.</i> acetate, methyl acetate, thiocyanate. See Tables IR-8.1 and IR-8.2 and Table IX for more examples of ‘ate’ anion names. See also ‘inate’, ‘onate’.
ato	Ending of name of any anion with an ‘ate’ name (see above) acting as a ligand, <i>e.g.</i> tetrahydridoaluminato(1–), nitrato, acetato. <i>Cf.</i> Sections IR-7.1.3 and IR-9.2.2.3 and Table IX. See also ‘inato’, ‘onato’. Ending of prefixes for certain anionic substituent groups, <i>e.g.</i> carboxylato, $-\text{C}(=\text{O})\text{O}^-$; phosphato, $-\text{O}-\text{P}(=\text{O})(\text{O}^-)_2$. See also ‘onato’.
diene	See ‘ene’.
diide	See ‘ide’.
diido	See ‘ido’.
dium	See ‘ium’.
diyl	Combined suffix composed of the suffix ‘yl’ and the multiplicative prefix ‘di’, indicating the loss of two hydrogen atoms from a parent hydride resulting in a diradical, or a substituent group with two single bonds, if necessary accompanied by locants, <i>e.g.</i> hydrazine-1,2-diyl, $\bullet\text{HNNH}\bullet$ or $-\text{HNNH}-$; phosphanediyli, $\text{HP}<$. See also ‘ylidene’.
diylum	See ‘ylum’.
ecane	Ending of parent names of ten-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.

Table III *Continued*

ecine	Ending of parent names of ten-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
ene	Ending of systematic names of acyclic and cyclic parent structures with double-bond unsaturation, replacing ‘ane’ in the name of the corresponding saturated parent hydride, and if necessary accompanied by locants and multiplicative prefixes specifying the locations and number of double bonds, <i>e.g.</i> diazene, triazene, pentasil-1-ene, cyclopenta-1,3-diene. <i>Cf.</i> Sections IR-6.2.2.3 and IR- 6.2.2.4. Ending of certain acceptable non-systematic names of unsaturated cyclic parent hydrides, <i>e.g.</i> benzene, azulene. See also ‘irene’, ‘ocene’.
enide	Combined ending of names of anions resulting from the removal of a hydron from a parent hydride with an ‘ene’ name, formed by adding the suffix ‘ide’, <i>e.g.</i> diazenide, $\text{HN}=\text{N}^-$. <i>Cf.</i> Section IR-6.4.4.
enium	Combined ending of names of cations resulting from the addition of a hydron to a parent structure with an ‘ene’ name, formed by adding the suffix ‘ium’, <i>e.g.</i> diazenium. <i>Cf.</i> Section IR-6.4.1. Combined ending resulting from the addition of the suffix ‘ium’ to a metallocene name. This leads to ambiguous names, see Section IR-10.2.6.
eno	Ending resulting from the change of the ‘ene’ ending in names of cyclic mancude ring systems to ‘eno’ to form prefixes in fusion nomenclature. (See Section P-25.3 of the Blue Book ^b .) Ending resulting from the change of the ‘ene’ ending in names of parent hydrides to form prefixes denoting bridging divalent substituent groups, <i>e.g.</i> diazeno, $-\text{N}=\text{N}-$.
epane	Ending of parent names of seven-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
epine	Ending of parent names of seven-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
etane	General ending of parent names of four-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3. See also ‘etidine’.
ete	Ending of parent names of four-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
etidine	Ending of parent names of four-membered nitrogen-containing saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
ic	Ending of names of many acids, both inorganic and organic, <i>e.g.</i> sulfuric acid, acetic acid, benzoic acid. For more examples, particularly of inorganic ‘ic’ acid names, see Tables IR-8.1 and IR-8.2, and Table IX. See also ‘inic’ and ‘onic’. Ending formerly added to stems of element names to indicate a higher oxidation state, <i>e.g.</i> ferric chloride, cupric oxide, ceric sulfate. Such names are no longer acceptable.
ide	Ending of names of monoatomic and homopolyatomic anions, <i>e.g.</i> chloride, sulfide, disulfide(2–), triiodide(1–). <i>Cf.</i> Sections IR-5.3.3.2 and IR-5.3.3.3 and Table IX. Ending of names of formally electronegative homoatomic constituents in compositional names, <i>e.g.</i> disulfur dichloride. <i>Cf.</i> Section IR-5.4.

Table III Continued

	Ending of some acceptable non-systematic names of heteropolyatomic anions: cyanide, hydroxide.
	Suffix for names of anions formed by removal of one or more hydrons from a parent hydride, accompanied by locants and multiplicative prefixes as appropriate, <i>e.g.</i> hydrazinide, H_2NNH^- ; hydrazine-1,2-diide, $^-\text{HNNH}^-$; disulfanediide, S_2^{2-} ; methanide, CH_3^- .
ido	Ending of name of any anion with an 'ide' name (see above) acting as a ligand, <i>e.g.</i> chlorido, disulfido(2-) or disulfanediido, hydrazinido, hydrazine-1,2-diido, methanido. <i>Cf.</i> Sections IR-7.1.3 and IR-9.2.2.3 and Table IX.
	Ending of certain prefixes for anionic substituent groups, <i>e.g.</i> oxido for $-\text{O}^-$.
inane	Ending of parent names of six-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
inate	Ending of names of anions and esters of 'inic' oxoacids, <i>e.g.</i> borinate, phosphinate.
inato	Modification of the 'inate' ending of an anion name (see above) used when the anion acts as a ligand.
ine	Ending of the non-systematic, but still acceptable, parent name hydrazine (N_2H_4) and of the now obsolete names of other Group 15 hydrides, <i>e.g.</i> phosphine (PH_3). Ending of names of large heteromonocycles (more than 10 ring atoms) with the maximum number of non-cumulative double bonds for use in fusion nomenclature, <i>e.g.</i> 2 <i>H</i> -1-oxa-4,8,11-triazacyclotetradecine. Last part of endings of a number of parent names in Hantzsch–Widman nomenclature of heteromonocycles, <i>i.e.</i> of 'irine', 'iridine', 'etidine', 'olidine', 'ine', 'inine', 'epine', 'ocine', 'onine' and 'ecine' (see Section IR-6.2.4.3). Ending of a number of parent names of nitrogenous heterocyclic parent hydrides, <i>e.g.</i> pyridine, acridine.
inic	Ending of the parent names of acids of the types $\text{H}_2\text{X}(=\text{O})(\text{OH})$ ($\text{X}=\text{N}, \text{P}, \text{As}, \text{Sb}$), <i>e.g.</i> stibinic acid; $\text{HX}(=\text{O})(\text{OH})$ ($\text{X}=\text{S}, \text{Se}, \text{Te}$), <i>e.g.</i> sulfinic acid; and of borinic acid, H_2BOH .
inide	Combined ending of names of anions resulting from the removal of a hydron from a parent hydride with an 'ine' name, formed by adding the suffix 'ide', <i>e.g.</i> hydrazinide, H_2NNH^- . <i>Cf.</i> Section IR-6.4.4.
inine	Ending of parent names of six-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
inite	Ending of names of anions and esters of oxoacids with an 'inous' name, <i>e.g.</i> phosphinite, H_2PO^- , from phosphinous acid.
inito	Modification of the 'inite' ending of an anion name (see above) used when the anion acts as a ligand.
inium	Combined ending of names of cations resulting from the addition of a hydron to a parent structure with an 'ine' name, formed by adding the suffix 'ium', <i>e.g.</i> hydrazinium, pyridinium. <i>Cf.</i> Section IR-6.4.1.
ino	Ending of some non-systematic substituent group prefixes, <i>e.g.</i> amino, NH_2- ; hydrazino, $\text{H}_2\text{NNH}-$. Ending resulting from the change of the 'ine' ending in names of cyclic mancude ring systems to 'ino' to form prefixes in fusion nomenclature. (See Section P-25.3 of the Blue Book ^b .)

Table III *Continued*

inous	Ending of the parent names of acids of the types $H_2X(OH)$ ($X = N, P, As, Sb$), <i>e.g.</i> stibinous acid. See Table IR-8.1 for other such names.
inoyl	Ending of prefixes for substituent groups formed by removing all hydroxy groups from 'inic' acids (see above), <i>e.g.</i> phosphinoyl, $H_2P(O)-$; seleninoyl, $HSe(O)-$. (See Table IR-8.1 for phosphinic and seleninic acids.)
inyl	Ending of prefixes for the divalent substituent groups $>X=O$ (sulfinyl, seleninyl and tellurinyl for $X = S, Se$ and Te , respectively).
io	Ending of acceptable alternative prefixes for certain cationic substituent groups, <i>e.g.</i> ammonio for azaniumyl, pyridinio for pyridiniumyl (<i>cf.</i> Section IR-6.4.9). Now abandoned ending of prefixes for substituent groups consisting of a single atom, <i>e.g.</i> mercurio, $-Hg-$.
irane	General ending of parent names of three-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3. See also 'iridine'.
irene	General ending of parent names of three-membered heteromonocycles with the maximum number of non-cumulative double bonds (<i>i.e.</i> one double bond) in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3. See also 'irine'.
iridine	Ending of parent names of three-membered nitrogen-containing saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
irine	Ending of parent names of three-membered heteromonocycles with the maximum number of non-cumulative double bonds (<i>i.e.</i> one double bond) and N as the only heteroatom(s) in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
ite	Ending of names of anions and esters of oxoacids having the 'ous' or the 'orous' ending in the acid name, <i>e.g.</i> hypochlorite (from hypochlorous acid), methyl sulfite (from sulfurous acid). <i>Cf.</i> Table IR-8.1. See also 'inite', 'onite'.
ito	Ending of name of any anion with an 'ite' name (see above) acting as a ligand, <i>e.g.</i> nitrito, sulfito. <i>Cf.</i> Sections IR-7.1.3 and IR-9.2.2.3 and Table IX. See also 'inito', 'onito'.
ium	Ending of names of many elements and their cations, <i>e.g.</i> helium, seaborgium, thallium(1+), and of the name of any new element (<i>cf.</i> Ref. 1 of Chapter IR-3). Suffix to indicate addition of hydrons to a parent hydride or other parent structure (see 'anium', 'enium', 'inium', 'onium', 'ynium'), accompanied by multiplying prefixes and locants as appropriate, <i>e.g.</i> hydrazinium, $H_2NNH_3^+$; hydrazine-1,2-dium, $^+H_3NNH_3^+$.
o	Terminal vowel indicating a negatively charged ligand; see 'ato', 'ido', 'ito'. Terminal vowel of prefixes for many inorganic and organic substituent groups, <i>e.g.</i> amino, chloro, oxido, sulfo, thiolato. Terminal vowel of prefixes for fusion components. (See Section P-25.3 of the Blue Book ^b .) See also 'eno', 'ino'. Terminal vowel of infixes used in functional replacement nomenclature (Section IR-8.6) to indicate replacement of oxygen atoms and/or hydroxy groups, <i>e.g.</i> 'amido', 'nitrido', 'thio'.
ocane	Ending of parent names of eight-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
ocene	Ending of the names of certain bis(cyclopentadienyl)metal compounds, <i>e.g.</i> ferrocene. <i>Cf.</i> Section IR-10.2.6.

Table III *Continued*

ocene	Ending of parent names of eight-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
ol	Suffix specifying substitution of a hydrogen atom in a parent hydride for the group –OH, accompanied by locants and multiplicative prefixes if appropriate, <i>e.g.</i> silanol, SiH ₃ OH; trisilane-1,3-diol, SiH ₂ (OH)SiH ₂ SiH ₂ OH. Ending of corresponding suffixes ‘thiol’, ‘selenol’, ‘tellurol’ for –SH, –SeH and –TeH, respectively.
olane	General ending of parent names of five-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> IR-6.2.4.3. See also ‘olidine’.
olate	Suffix specifying substitution of a hydrogen atom in a parent hydride for the substituent –O [–] , accompanied by locants and multiplicative prefixes if appropriate, <i>e.g.</i> silanolate, SiH ₃ O [–] ; trisilane-1,3-diolate, SiH ₂ (O [–])SiH ₂ SiH ₂ O [–] . Ending of corresponding suffixes ‘thiolate’, ‘selenolate’, ‘tellurolate’ for –S [–] , –Se [–] and –Te [–] , respectively.
olato	Modification of the suffix ‘olate’ used when the anion in question acts as a ligand.
ole	Ending of parent names of five-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
olidine	Ending of parent names of five-membered nitrogen-containing saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
onane	Ending of parent names of nine-membered saturated heteromonocycles in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
onate	Ending of names of anions and esters of ‘onic’ oxoacids, <i>e.g.</i> boronate, phosphonate, tetrathionate.
onato	Modification of the ‘onate’ ending of an anion used when the anion acts as a ligand. Ending of prefixes of certain anionic substituent groups, <i>e.g.</i> phosphonato, –P(=O)(O [–]) ₂ ; sulfonato, –S(=O) ₂ (O [–]).
one	Suffix specifying the substitution of two hydrogen atoms on the same skeletal atom in a parent hydride for the substituent =O, accompanied by locants and multiplicative prefixes as appropriate, <i>e.g.</i> phosphanone, HP=O; pentane-2,4-dione, CH ₃ C(=O)CH ₂ C(=O)CH ₃ . Ending of corresponding suffixes ‘thione’, ‘selenone’, ‘tellurone’ for =S, =Se and =Te, respectively.
onic	Ending of the parent names of acids of the types HXO(OH) ₂ (X=N, P, As, Sb), <i>e.g.</i> stibonic acid; HXO ₂ (OH) (X=S, Se, Te), <i>e.g.</i> sulfonic acid; and of boronic acid, HB(OH) ₂ . See Table IR-8.1. Ending of the parent names dithionic, trithionic, <i>etc.</i> , acids (see Table IR-8.1).
onine	Ending of parent names of nine-membered heteromonocycles with the maximum number of non-cumulative double bonds in Hantzsch–Widman nomenclature, <i>cf.</i> Section IR-6.2.4.3.
onite	Ending of names of anions and esters of ‘onous’ oxoacids, <i>e.g.</i> phosphonite, tetrathionite.
onito	Modification of the ‘onite’ ending of an anion name used when the anion acts as a ligand.

Table III *Continued*

onium	Ending of still acceptable non-systematic names of cations formed by hydron addition to a mononuclear parent hydride: ammonium, oxonium (see Section IR-6.4.1).
ono	Ending of prefixes for substituent groups formed from 'onic' acids by removal of a hydrogen atom, <i>e.g.</i> phosphono for --P(=O)(OH)_2 . <i>Exception:</i> note that $\text{--S(=O)}_2\text{OH}$ is 'sulfo' rather than 'sulfono'.
onous	Ending of the parent names of acids of the types HX(OH)_2 ($\text{X} = \text{N, P, As, Sb}$), <i>e.g.</i> stibonous acid. Ending of the parent names dithionous, trithionous, <i>etc.</i> , acids (see Table IR-8.1).
onoyl	Ending of prefixes for substituent groups formed by removing all hydroxy groups from 'onic' acids, <i>e.g.</i> phosphonoyl, HP(O)< ; selenonoyl, $\text{HSe(O)}_2\text{--}$. (See Table IR-8.1 for phosphonic and selenonic acids.)
onyl	Ending of prefixes for the divalent substituent groups >X(=O)_2 (sulfonyl, selenonyl and telluronyl for $\text{X} = \text{S, Se and Te}$, respectively).
orane	Ending of the acceptable alternative names phosphorane for λ^5 -phosphane (PH_5), arsorane for λ^5 -arsane (AsH_5) and stiborane for λ^5 -stibane (SbH_5).
oryl	Ending of prefixes for substituent groups formed by removing all hydroxy groups from 'oric' acids, <i>e.g.</i> phosphoryl, P(O)< , from phosphoric acid.
ous	Ending of parent names of certain inorganic oxoacids, <i>e.g.</i> arsorous acid, seleninous acid. For more examples of 'ous' acid names, see Tables IR-8.1 and IR-8.2. See also 'inous', 'onous'. Ending formerly added to stems of element names to indicate a lower oxidation state, <i>e.g.</i> ferrous chloride, cuprous oxide, cerous hydroxide. Such names are no longer acceptable.
triene	See 'ene'.
triide	See 'ide'.
trium	See 'ium'.
triyl	Combined suffix composed of the suffix 'yl' and the multiplying prefix 'tri', indicating the loss of three hydrogen atoms from a parent hydride resulting in a triradical or a substituent group forming three single bonds, <i>e.g.</i> the substituent groups boranetriyl, --B< ; trisilane-1,2,3-triyl, $\text{--SiH}_2\text{SiHSiH}_2\text{--}$; λ^5 -phosphanetriyl, $\text{H}_2\text{P<}$. (See also 'ylidyne' and 'ylylidene'.)
uide	Suffix specifying the addition of hydride to a parent structure, accompanied by locants and multiplicative prefixes if appropriate, <i>e.g.</i> tellanuide, TeH_3^- .
uido	Modification of the 'uide' suffix in an anion name used when the anion acts as a ligand.
y	Terminal vowel of prefixes for some substituent groups, <i>e.g.</i> carboxy, --COOH ; hydroxy, --OH ; oxy, --O-- . Terminal vowel in prefixes used in specifying chain and ring atoms in additive nomenclature for inorganic chains and rings, <i>cf.</i> Section IR-7.4. These prefixes are given for all elements in Table X.
yl	Suffix to indicate removal of hydrogen atoms from a parent hydride to form radicals or substituent groups, accompanied by multiplicative prefixes and locants as appropriate, <i>e.g.</i> hydrazinyl, $\text{H}_2\text{NNH}^\bullet$ or $\text{H}_2\text{NNH--}$; hydrazine-1,2-diyl, $\bullet\text{HNNH}^\bullet$ or --HNNH-- . (See also 'diyl', 'ylene', 'ylidene', 'triyl', 'ylylidene', 'ylidyne'.)

Table III *Continued*

	Ending of certain non-systematic names of oxidometal cations, <i>e.g.</i> vanadyl for oxidovanadium(2+). These names are no longer acceptable.
ylene	Ending of a few still acceptable names for divalent substituent groups, meaning the same as 'diyl': methylene for methanediyl, $-\text{CH}_2-$; phenylene for benzenediyl, $-\text{C}_6\text{H}_4-$; (1,2-phenylene for benzene-1,2-diyl <i>etc.</i>).
ylidene	Suffix for names of divalent substituent groups formed by the loss of two hydrogen atoms from the same atom of a parent hydride and forming a double bond, <i>e.g.</i> azanylidene, $\text{HN}=\text{}$, and for names of corresponding diradicals. (See also 'diyl'.)
ylidyne	Suffix for names of trivalent substituent groups formed by the loss of three hydrogen atoms from the same atom of a parent hydride and forming a triple bond, <i>e.g.</i> phosphanylidyne, $\text{P}\equiv$. (See also 'ylylidene' and 'triyl'.)
ylum	Suffix for names of cations formed by the loss of hydride ions from parent hydrides, accompanied by locants and multiplicative prefixes as appropriate, <i>e.g.</i> azanylium, NH_2^+ ; disilane-1,2-dylium, $^+\text{H}_2\text{SiSiH}_2^+$.
ylylidene	Combined suffix ('yl' plus 'ylidene') for names of trivalent substituent groups formed by the loss of three hydrogen atoms from the same atom, forming a single bond and a double bond, <i>e.g.</i> azanylylidene, $-\text{N}=\text{}$. (See also 'ylidyne' and 'triyl'.)
yne	Ending of systematic names of acyclic and cyclic parent structures with triple-bond unsaturation, replacing 'ane' in the name of the corresponding saturated parent hydride, and if necessary accompanied by locants and multiplicative prefixes specifying the locations and number of triple bonds, <i>e.g.</i> diazyne (see 'ynium' for an application of this name), ethyne, penta-1,4-diyne.
ynide	Combined ending of names of anions resulting from the removal of a hydron from a parent hydride with an 'yne' name, formed by adding the suffix 'ide', <i>e.g.</i> ethynide, $\text{CH}\equiv\text{C}^-$. <i>Cf.</i> Section IR-6.4.4.
ynium	Combined ending of names of cations resulting from the addition of a hydron to a parent structure with an 'yne' name, formed by adding the suffix 'ium', <i>e.g.</i> diazynium ($\text{N}\equiv\text{NH}^+$). <i>Cf.</i> Section IR-6.4.1.

^a The term 'suffix' is understood here to mean a name part added to a parent name in order to specify a modification of that parent, *e.g.* substitution of a hydrogen atom in a parent hydride by a characteristic group (suffixes such as 'carboxylic acid', 'thiol', *etc.*) or formation of a radical or substituent group by removal of one or more hydrogen atoms (suffixes such as 'yl', 'ylidene', *etc.*). The term 'ending' is used in a broader sense, but also to designate specifically the common last part (last syllable or last few syllables) of systematic names for members of classes of compounds (such as 'ane', 'ene', 'diene', 'yne', *etc.*, for parent hydrides, and 'onic acid', 'inic acid', *etc.*, for inorganic oxoacids).

^b *Nomenclature of Organic Chemistry, IUPAC Recommendations*, eds. W.H. Powell and H. Favre, Royal Society of Chemistry, in preparation. (The Blue Book.)

Table IV *Multiplicative prefixes*

1	mono	21	henicosa
2	di ^a (bis ^b)	22	docosa
3	tri (tris)	23	tricos
4	tetra (tetrakis)	30	triaconta
5	penta (pentakis)	31	hentriaconta
6	hexa (hexakis)	35	pentatriaconta
7	hepta (heptakis)	40	tetraconta
8	octa (octakis)	48	octatetraconta
9	nona (nonakis)	50	pentaconta
10	deca (decakis)	52	dopentaconta
11	undeca	60	hexaconta
12	dodeca	70	heptaconta
13	trideca	80	octaconta
14	tetradeca	90	nonaconta
15	pentadeca	100	hecta
16	hexadeca	200	dicta
17	heptadeca	500	pentacta
18	octadeca	1000	kilia
19	nonadeca	2000	dilia
20	icosa		

^a In the case of a ligand using two donor atoms, the term 'bidentate' rather than 'didentate' is recommended because of prevailing usage.

^b The prefixes bis, tris, *etc.* (examples are given for 1–10 but continue throughout) are used with composite ligand names or in order to avoid ambiguity.

Table V *Geometrical and structural affixes*

Except for those denoted by Greek letters, geometrical and structural affixes are italicized. All are separated from the rest of the name by hyphens.

<i>antiprisma</i>	eight atoms bound into a regular antiprism
<i>arachno</i>	a boron structure intermediate between <i>nido</i> and <i>hypho</i> in degree of openness
<i>asym</i>	asymmetrical
<i>catena</i>	a chain structure; often used to designate linear polymeric substances
<i>cis</i>	two groups occupying adjacent positions in a coordination sphere
<i>closo</i>	a cage or closed structure, especially a boron skeleton that is a polyhedron having all faces triangular
<i>cyclo</i>	a ring structure. (Here, <i>cyclo</i> is used as a modifier indicating structure and hence is italicized. In organic nomenclature, 'cyclo' is considered to be part of the parent name since it changes the molecular formula. It is therefore not italicized.)
δ (delta)	denotes the absolute configuration of chelate ring conformations
Δ (delta)	a structural descriptor to designate deltahedra, or shows absolute configuration
<i>dodecahedro</i>	eight atoms bound into a dodecahedron with triangular faces
η (eta)	specifies the bonding of contiguous atoms of a ligand to a central atom
<i>fac</i>	three groups occupying the corners of the same face of an octahedron
<i>hexahedro</i>	eight atoms bound into a hexahedron (<i>e.g.</i> cube)
<i>hexaprisma</i>	twelve atoms bound into a hexagonal prism
<i>hypho</i>	an open structure, especially a boron skeleton, more closed than a <i>klado</i> structure but more open than an <i>arachno</i> structure
<i>icosahedro</i>	twelve atoms bound into an icosahedron with triangular faces
κ (kappa)	specifies the donor atoms in a ligand
<i>klado</i>	a very open polyboron structure
λ (lambda)	signifies, with its superscript, the bonding number, <i>i.e.</i> the sum of the number of skeletal bonds and the number of hydrogen atoms associated with an atom in a parent compound; denotes the absolute configuration of chelate ring conformations
Λ (lambda)	shows absolute configuration
<i>mer</i>	meridional; three groups occupying vertices of an octahedron so that one is <i>cis</i> to the other two which are themselves mutually <i>trans</i>
μ (mu)	signifies that a group so designated bridges two or more coordination centres
<i>nido</i>	a nest-like structure, especially a boron skeleton that is almost closed
<i>octahedro</i>	six atoms bound into an octahedron
<i>pentaprisma</i>	ten atoms bound into a pentagonal prism
<i>quadro</i>	four atoms bound into a quadrangle (<i>e.g.</i> square)
<i>sym</i>	symmetrical
<i>tetrahedro</i>	four atoms bound into a tetrahedron
<i>trans</i>	two groups occupying positions in a coordination sphere directly opposite each other
<i>triangulo</i>	three atoms bound into a triangle
<i>triprisma</i>	six atoms bound into a triangular prism

Table VI *Element sequence*

→ He	Ne	Ar	Kr	Xe	Rn	→ Li	Na	K	Rb	Cs	Fr	→ Be	Mg	Ca	Sr	Ba	Ra	→ Sc	Y	La	Lu	→ Ti	Zr	Hf	Rf	→ V	Nb	Ta	Db	→ Cr	Mo	W	Sg	→ Mn	Tc	Re	Bh	→ Fe	Ru	Os	Hs	→ Co	Rh	Ir	Mt	→ Ni	Pd	Pt	Ds	→ Cu	Ag	Au	Rg	→ Zn	Cd	Hg	→ B	Al	Ga	In	Tl	→ C	Si	Ge	Sn	Pb	→ N	P	As	Sb	Bi	→ O	S	Se	Te	Po	→ F	Cl	Br	I	At
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Table VII Ligand abbreviations

Guidelines for the construction and use of ligand abbreviations are given in Section IR-4.4.4 and their use in the formulae of coordination complexes is described in Section IR-9.2.3.4. Abbreviations are listed in alphabetical order but those beginning with a numeral are listed by the first letter of the abbreviation (e.g. 2,3,2-tet appears under the letter ‘t’).

Structural formulae of selected ligands are shown in Table VIII (numbered according to the present Table).

<i>Number and abbreviation</i>	<i>Systematic name</i>	<i>Other name (from which abbreviation derived)^a</i>
1. 4-abu	4-aminobutanoato	
2. Ac	acetyl	
3. acac	2,4-dioxopentan-3-ido	acetylacetonato
4. acacen	2,2'-[ethane-1,2-diylbis(azanylylidene)]bis(4-oxopentan-3-ido)	bis(acetylacetonato)ethylenediamine
5. ade	9H-purin-6-amine	adenine
6. ado	9-β-D-ribofuranosyl-9H-purin-6-amine	adenosine
7. adp	adenosine 5'-diphosphato(3-)	
8. aet	2-aminoethanethiolato	
9. ala	2-aminopropanoato	alaninato
10. ama	2-aminopropanedioato	aminomalonato
11. amp	adenosine 5'-phosphato(2-)	adenosine monophosphato
12. [9]aneN ₃ (also taen)	1,4,7-triazonane	
13. [12]aneN ₄ (also cyclen)	1,4,7,10-tetraazacyclododecane	
14. [14]aneN ₄ (also cyclam)	1,4,8,11-tetraazacyclotetradecane	
15. [18]aneP ₄ O ₂	1,10-dioxo-4,7,13,16-tetraphosphacyclooctadecane	
16. [9]aneS ₃	1,4,7-trithionane	
17. [12]aneS ₄	1,4,7,10-tetrathiaacyclododecane	
18. arg	2-amino-5-carbamimidamidopentanoato	argininato
19. asn	2,4-diamino-4-oxobutanoato	asparaginato
20. asp	2-aminobutanedioato	aspartato
21. atmp	[nitrilotris(methylene)]tris(phosphonato)	aminotris(methylenephosphonato)
22. atp	adenosine 5'-triphosphato(4-)	
23. 2,3-bdta	2,2',2'',2'''-(butane-2,3-diylidinitrilo)tetraacetato	

Table VII Continued

Number and abbreviation	Systematic name	Other name (from which abbreviation derived) ^a
24. benzo-15-crown-5	2,3,5,6,8,9,11,12-octahydro-1,4,7,10,13-benzopentaoxacyclopentadecine	
25. big	bis(carbamimidoyl)azanido	biguanid-3-ido
26. biim	2,2'-bi(1 <i>H</i> -imidazole)-1,1'-diido	2,2'-biimidazolato
27. binap	1,1'-binaphthalene-2,2'-diylbis(diphenylphosphane)	
28. bn	butane-2,3-diamine	
29. bpy	2,2'-bipyridine	
30. 4,4'-bpy	4,4'-bipyridine	
31. Bu	butyl	
32. bzac	1,3-dioxo-1-phenylbutan-2-ido	benzoylacetato
33. bzim	1 <i>H</i> -benzimidazol-1-ido	
34. Bz ^b	benzyl	
35. bztz	1,3-benzothiazole	
36. cat	benzene-1,2-diolato	catecholato
37. cbdea	cyclobutane-1,1-dicarboxylato	
38. cdta	2,2',2'',2'''-(cyclohexane-1,2-diylidinitrilo)tetraacetato	
39. C ₅ H ₄ Me	methylcyclopentadienyl	
40. chxn (also dach)	cyclohexane-1,2-diamine	
41. cit	2-hydroxypropane-1,2,3-tricarboxylato	cittrato
42. C ₅ Me ₅ ^c	pentamethylcyclopentadienyl	
43. cod	cycloocta-1,5-diene	
44. cot	cycloocta-1,3,5,7-tetraene	
45. Cp	cyclopentadienyl	
46. cptn	cyclopentane-1,2-diamine	
47. 18-crown-6	1,4,7,10,13,16-hexaoxacyclooctadecane	
48. crypt-211	4,7,13,18-tetraoxa-1,10-diazabicyclo[8.5.5]icosane	cryptand 211
49. crypt-222	4,7,13,16,21,24-hexaoxa-1,10-diazabicyclo[8.8.8]hexacosane	cryptand 222
50. Cy	cyclohexyl	

cyclam (see [14]aneN ₄ , No. 14)		
cyclen (see [12]aneN ₄ , No. 13)		
51. cys	2-amino-3-sulfanypropanoato	cysteinato
52. cyt	4-aminopyrimidin-2(1 <i>H</i>)-one	cytosine
53. dabco	1,4-diazabicyclo[2.2.2]octane	
dach (see chxn, No. 40)		
54. dbm	1,3-dioxo-1,3-diphenylpropan-2-ido	diaminocyclohexane
55. dea	2,2'-azanediyldi(ethan-1-olato)	dibenzoylmethanato
56. depe	ethane-1,2-diylbis(diethylphosphane)	diethanolaminato
57. diars	benzene-1,2-diylbis(dimethylarsane)	1,2-bis(diethylphosphino)ethane
58. dien	<i>N</i> -(2-aminoethyl)ethane-1,2-diamine	diethylenetriamine
59. [14]1,3-dieneN ₄	1,4,8,11-tetraazacyclotetradeca-1,3-diene	
60. diop	[(2,2-dimethyl-1,3-dioxolane-4,5-diyl)bis(methylene)]bis(diphenylphosphane)	
61. diox	1,4-dioxane	
62. dipamp	ethane-1,2-diylbis[(2-methoxyphenyl)phenylphosphane]	'dimer of phenylanisylmethylphosphine'
63. dma	<i>N,N</i> -dimethylacetamide	dimethylacetamide
64. dme	1,2-dimethoxyethane	
65. dmf	<i>N,N</i> -dimethylformamide	
66. dmg	butane-2,3-diylidenebis(azanolato)	dimethylglyoximate
67. dmpe	ethane-1,2-diylbis(dimethylphosphane)	1,2-bis(dimethylphosphino)ethane
68. dmpm	methylenebis(dimethylphosphane)	bis(dimethylphosphino)methane
69. dmso	(methanesulfinyl)methane	dimethyl sulfoxide
70. dpm	2,2,6,6-tetramethyl-3,5-dioxoheptan-4-ido	dipivaloylmethanato
71. dppe	ethane-1,2-diylbis(diphenylphosphane)	1,2-bis(diphenylphosphino)ethane
72. dppl	1,1'-bis(diphenylphosphanyl)ferrocene	
73. dppm	methylenebis(diphenylphosphane)	bis(diphenylphosphino)methane
74. dppp	propane-1,3-diylbis(diphenylphosphane)	1,3-bis(diphenylphosphino)propane
75. dtmpe	(phosphonomethyl)azanediylbis[ethane-2,1-diyl]nitrilobis(methylene)]tetrakis(phosphonato)	diethylenetriaminepentakis- C (methylene)phosphonato) ^d
76. dtpe	2,2',2''/2'''-(carboxylatomethyl)azanediylbis[ethane-2,1-diyl]nitriolo]tetracetato	diethylenetriaminepentaacetato
77. ea	2-amino(ethan-1-olato)	ethanolaminato

Table VII Continued

Number and abbreviation	Systematic name	Other name (from which abbreviation derived) ^a
78. edda	2,2'-(ethane-1,2-diylbis(azanediyl))diacetato	ethylenediaminediacetato
79. edta	2,2',2'',2'''-(ethane-1,2-diyl(dinitrilo)tetraacetato	ethylenediaminetetraacetato
80. edtmpa	ethane-1,2-diylbis[nitrilobis(methylene)]tetrakis(phosphonato)	ethylenediaminetetrakis(methylenephosphonato) ^d
81. egta	2,2',2'',2'''-(ethane-1,2-diylbis(oxyethane-2,1-diyl(nitrilo))tetraacetato	ethylene glycol-bis(2-aminoethyl)- <i>N,N,N',N'</i> -tetraacetic acid
82. en	ethane-1,2-diamine	
83. Et	ethyl	
84. Et ₂ dic	<i>N,N</i> -diethylcarbomodithioato	<i>N,N</i> -diethyldithiocarbamato
85. fod	6,6,7,7,8,8,8-heptafluoro-2,2-dimethyl-3,5-dioxooctan-4-ido	
86. fta	1,1,1-trifluoro-2,4-dioxopentan-3-ido	trifluoroacetylacetonato
87. gln	2,5-diamino-5-oxopentanoato	glutaminato
88. glu	2-aminopentanedioato	glutamato
89. gly	aminoacetato	glycinato
90. gua	2-amino-9 <i>H</i> -purin-6(1 <i>H</i>)-one	guanine
91. guo	2-amino-9-β-D-ribofuranosyl-9 <i>H</i> -purin-6(1 <i>H</i>)-one	guanosine
92. hdtmpa	hexane-1,6-diylbis[nitrilobis(methylene)]tetrakis(phosphonato)	hexamethylenediaminetetrakis(methylenephosphonato) ^d
93. hedp	1-hydroxyethane-1,1-diylbis(phosphonato)	1-hydroxyethane-1,1-diphosphonato
94. hfa	1,1,1,5,5,5-hexafluoropentane-2,4-dioxopentan-3-ido	hexafluoroacetylacetonato
95. his	2-amino-3-(imidazol-4-yl)propanoato	histidinato
96. hmpa	hexamethylphosphoric triamide	
97. hmta	1,3,5,7-tetraazatricyclo[3.3.1.1 ^{3,7}]decane	hexamethylenetetramine
98. ida	2,2'-azanediylidiacetato	iminodiacetato
99. ile	2-amino-3-methylpentanoato	isoleucinato
100. im	1 <i>H</i> -imidazol-1-ido	
101. isn	pyridine-4-carboxamide	isonicotinamide

102. leu	2-amino-4-methylpentanoato	leucinato
103. lut	2,6-dimethylpyridine	lutidine
104. lys	2,6-diaminohexanoato	lysinato
105. mal	2-hydroxybutanedioato	malato
106. male	(Z)-butenedioato	maleato
107. malo	propanedioato	malonato
108. Me	methyl	
109. 2-Mepy	2-methylpyridine	
110. met	2-amino-4-(methylsulfonyl)butanoato	methioninato
111. mnt	1,2-dicyanoethene-1,2-dithiolato	maleonitriledithiolato
112. napy	1,8-naphthyridine	
113. nbd	bicyclo[2.2.1]hepta-2,5-diene	norbornadiene
114. nia	pyridine-3-carboxamide	nicotinamide
115. nmp	N-methylpyrrolidine	
116. nta	2,2',2''-nitrioltriacetato	
117. oep	2,3,7,8,12,13,17,18-octaethylporphyrin-21,23-diido	
118. ox	ethanedioato	oxalato
119. pc	phthalocyanine-29,31-diido	
120. 1,2-pdta	2,2',2'',2'''-(propane-1,2-diyl(dinitrilo))tetraacetato	1,2-propylenediaminetetraacetato
121. 1,3-pdta	2,2',2'',2'''-(propane-1,3-diyl(dinitrilo))tetraacetato	1,3-propylenediaminetetraacetato
122. Ph	phenyl	
123. phe	2-amino-3-phenylpropanoato	phenylalaninato
124. phen	1,10-phenanthroline	
125. pip	piperidine	
126. pmdien	2,2'-(methylazanediyl)bis(N,N-dimethylethan-1-amine)	N,N,N',N'',N'''-pentamethyl- C -diethylenetriamine ^d
127. pn	propane-1,2-diamine	
128. pplX	2,18-bis(2-carboxyethyl)-3,7,12,17-tetramethyl-8,13-divinylporphyrin-21,23-diido	protoporphyrinato IX
129. pro	pyrrolidine-2-carboxylato	prolinato
130. ptn	pentane-2,4-diamine	
131. py	pyridine	

Table VII Continued

Number and abbreviation	Systematic name	Other name (from which abbreviation derived) ^a
132. pyz	pyrazine	
133. pz	1 <i>H</i> -pyrazol-1-ido	
134. qdt	quinoxaline-2,3-dithiolato	
135. quin	quinolin-8-olato	
136. sal	2-hydroxybenzoato	salicylato
137. salan	2-[(phenylimino)methyl]phenolato	salicylideneanilinato
138. saldien	2,2'-[azanediy]bis(ethane-2,1-diylazanylylidenemethanylylidene)]diphenolato	bis(salicylidene)diethylenetriaminato
139. salen	2,2'-[ethane-1,2-diylbis(azanylylidenemethanylylidene)]diphenolato	bis(salicylidene)ethylenediaminato
140. salgly	<i>N</i> -(2-oxidobenzylidene)glycinato	salicylidene-glycinato
141. salpn	2,2'-[propane-1,2-diylbis(azanylylidenemethanylylidene)]diphenolato	bis(salicylidene)propylenediaminato
142. saltn	2,2'-[propane-1,3-diylbis(azanylylidenemethanylylidene)]diphenolato	bis(salicylidene)trimethylenediaminato
143. sdta ^e	2,2',2'',2'''-[(1,2-diphenylethane-1,2-diyl)dinitrilo]tetraacetato	stilbenediaminetetraacetato
144. sepf ^f	1,3,6,8,10,13,16,19-octaazabicyclo[6.6.6]icosane	
145. ser	2-amino-3-hydroxypropanoato	serinato
146. stien ^e	1,2-diphenylethane-1,2-diamine	stilbenediamine
tacn (see [9]aneN ₃ , No. 12)		1,4,7-triazacyclononane
147. tap	propane-1,2,3-triamine	1,2,3-triaminopropane
148. tart	2,3-dihydroxybutanedioato	tartrato
149. tcne	ethenetetracarbonitrile	tetracyanoethylene
150. tenq	2,2'-(cyclohexa-2,5-diene-1,4-diylidene)di(propanedinitrile)	tetracyanoquinodimethane
151. tdt	4-methylbenzene-1,2-dithiolato	
152. tea	2,2',2''-nitrioltri(ethan-1-olato)	triethanolaminato
153. terpy	2,2':6':2''-terpyridine	terpyridine
154. 2,3,2-tet	<i>N,N'</i> -bis(2-aminoethyl)propane-1,3-diamine	1,4,8,11-tetrazaundecane
155. 3,3,3-tet	<i>N,N'</i> -bis(3-aminopropyl)propane-1,3-diamine	1,5,9,13-tetraza-tridecane
156. tetren	<i>N,N'</i> -(azanediy)diethane-2,1-diyl)di(ethane-1,2-diamine)	tetraethylenepentamine

157. tfa	trifluoroacetato	
158. thf	oxolane	tetrahydrofuran
159. thiox	1,4-oxathiane	thioxane
160. thr	2-amino-3-hydroxybutanoato	threoninato
161. tht	thiolane	tetrahydrothiophene
162. thy	5-methylpyrimidine-2,4(1 <i>H</i> ,3 <i>H</i>)-dione	thymine
163. tmen	<i>N,N,N',N'</i> -tetramethylethane-1,2-diamine	
164. tmp	5,10,15,20-tetrakis(2,4,6-trimethylphenyl)porphyrin-21,23-diido	5,10,15,20-tetramesitylporphyrin-21,23-diido
165. tn	propane-1,3-diamine	trimethylenediamine
166. Tol (<i>o</i> -, <i>m</i> - or <i>p</i> -)	2-, 3- or 4-methylphenyl	tolyl (<i>o</i> -, <i>m</i> - or <i>p</i> -)
167. Tp	hydridotris(pyrazolido- <i>N</i>)borato(1-), or tris(1 <i>H</i> -pyrazol-1-yl)boranuido	hydrotis(pyrazolyl)borato
168. Tp ^g	tris(3,5-dimethylpyrazolido- <i>N</i>)hydridoborato(1-)	hydrotis(3,5-dimethylpyrazolyl)borato
169. tpp	5,10,15,20-tetraphenylporphyrin-21,23-diido	
170. tren	<i>N,N</i> -bis(2-aminoethyl)ethane-1,2-diamine	tris(2-aminoethyl)amine
171. trien	<i>N,N'</i> -bis(2-aminoethyl)ethane-1,2-diamine	triethylenetetramine
172. triphos ^h	[(phenylphosphanediy)bis(ethane-2,1-diyl)]bis(diphenylphosphane)	
173. tris	2-amino-2-(hydroxymethyl)propane-1,3-diol	
174. ttp	2-amino-3-(1 <i>H</i> -indol-3-yl)propanoato	tris(hydroxymethyl)aminomethane
175. tsalen	2,2'-[ethane-1,2-diyl]bis(azanylylidenemethanylylidene)]dibenzeneethiolato	tryptophanato
176. tffa	4,4,4-trifluoro-1,3-dioxo-1-(2-thienyl)butan-2-ido	bis(thiosalicylidene)ethylenediaminato
177. ttha	2,2',2'',2'''-(ethane-1,2-diyl)bis{[(carboxylatomethyl)azanediy]ethane-2,1-diylinitrilo}tetraacetato	thenoyltrifluoroacetato
178. ttp	5,10,15,20-tetrakis(4-methylphenyl)porphyrin-21,23-diido	triethylenetetraminehexaacetato
179. tu	thiourea	5,10,15,20-tetra- <i>p</i> -tolylporphyrin-21,23-diido
180. tyr	2-amino-3-(4-hydroxyphenyl)propanoato	tyrosinato
181. tz	1,3-thiazole	thiazole
182. ura	pyrimidine-2,4(1 <i>H</i> ,3 <i>H</i>)-dione	uracil
183. val	2-amino-3-methylbutanoato	valinato

^a Many of these names are no longer acceptable.

^b The abbreviation Bz has often been used previously for 'benzoyl', and Bzl has been used for 'benzyl'. Use of the alternatives, PhCO and PhCH₂, is therefore preferable.

^c The use of the abbreviation Cp* for pentamethylcyclopentadienyl is discouraged. It can lead to confusion because the asterisk, *, is also used to represent an excited state, an optically active substance, etc.

- ^d The symbol ‘ $\frown$ ’ is used to divide the name, necessitated by the line break. In the absence of the line break this symbol is omitted. Note that all *hyphens* are true parts of the name.
- ^e The abbreviation derives from the non-systematic name stilbenediamine which incorrectly implies the presence of a C=C double bond in the ligand.
- ^f The abbreviation derives from the non-systematic name sepulchrate which incorrectly implies that the ligand is anionic.
- ^g The use of Tp’ is preferred to Tp* for the reasons given in footnote c. A general procedure for abbreviating substituted hydridotris(pyrazolido-*N*)borate ligands has been proposed [see S. Trofimenko, *Chem. Rev.*, **93**, 943–980 (1993)]. For example, Tp’ becomes Tp^{Me₂}, the superscript denoting the methyl groups at the 3- and 5-positions of the pyrazole rings.
- ^h The abbreviation triphos should not be used for the four-phosphorus ligand PhP(CH₂PPh₂)₃.

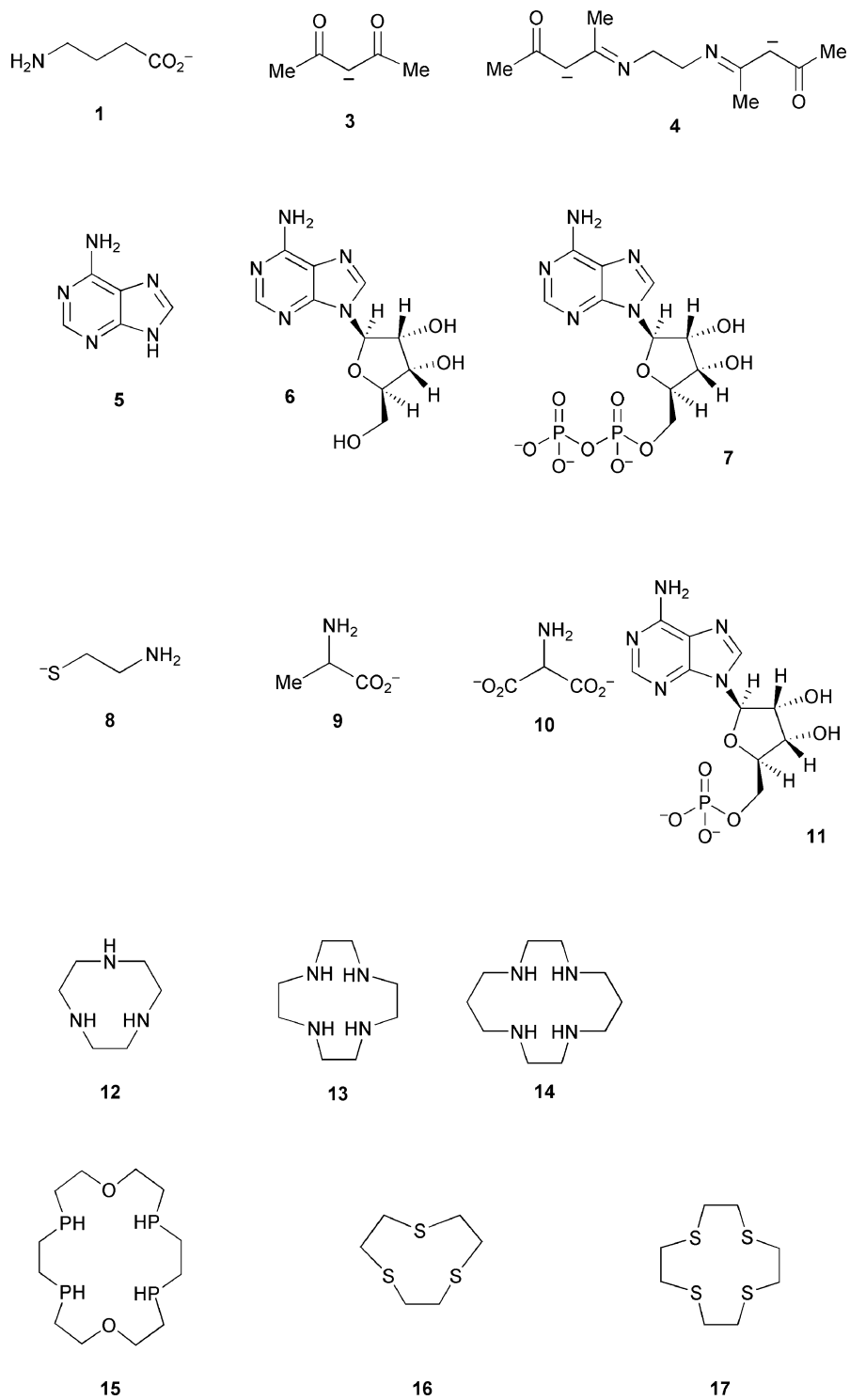
Table VIII Structural formulae of selected ligands (numbered according to Table VII)

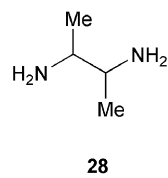
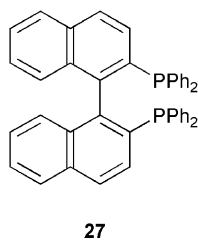
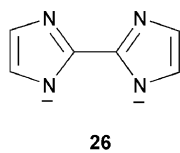
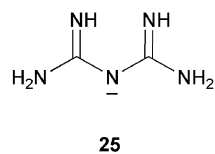
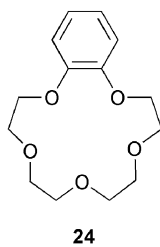
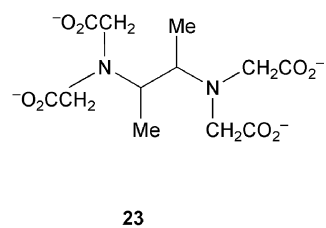
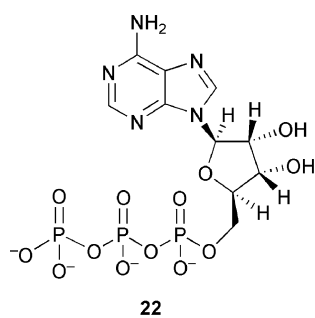
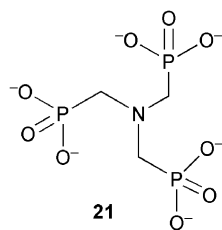
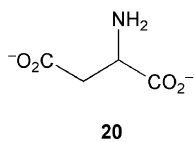
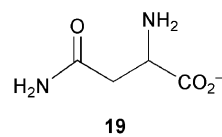
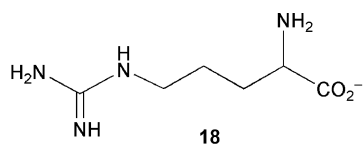
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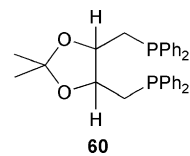
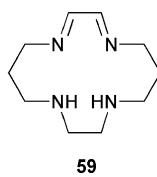
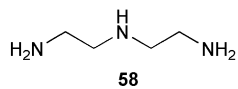
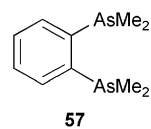
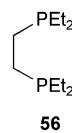
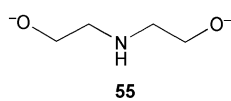
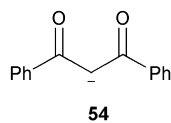
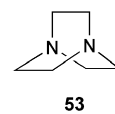
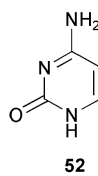
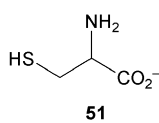
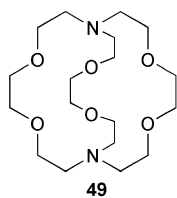
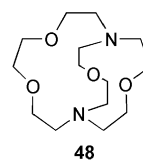
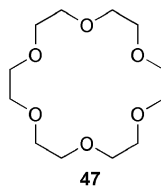
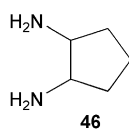
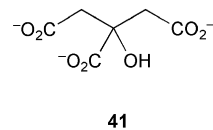
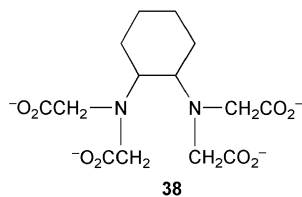
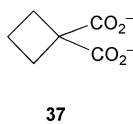
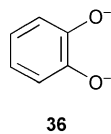
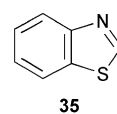
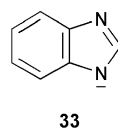
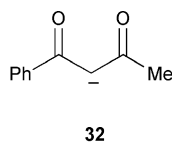
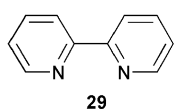
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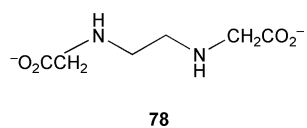
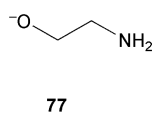
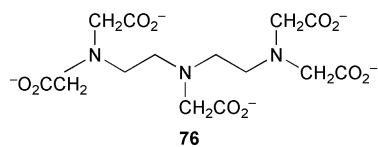
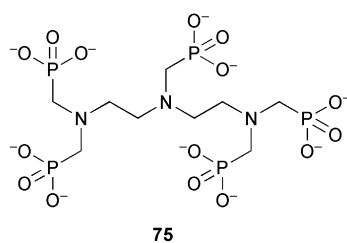
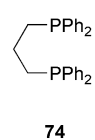
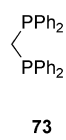
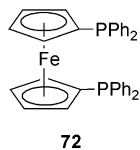
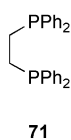
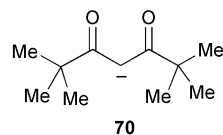
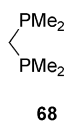
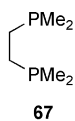
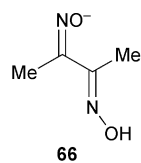
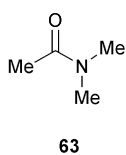
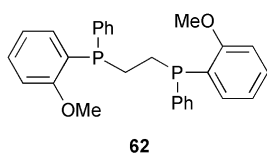
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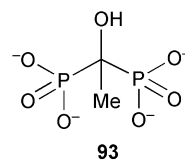
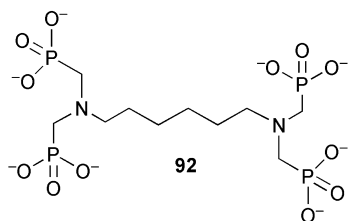
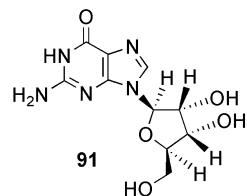
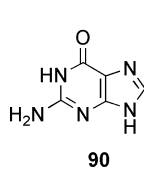
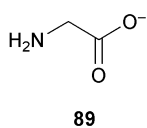
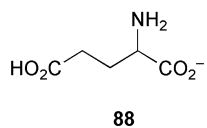
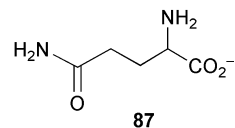
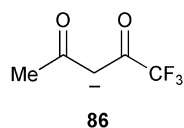
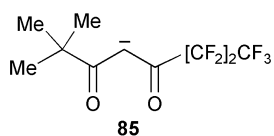
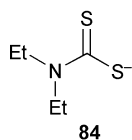
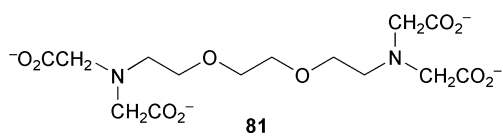
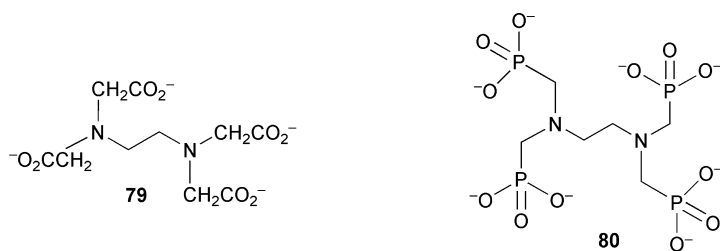


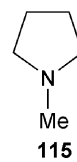
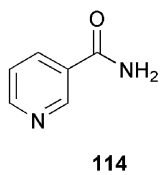
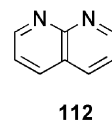
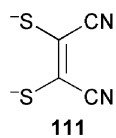
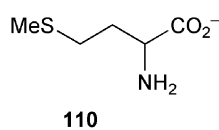
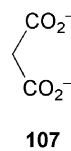
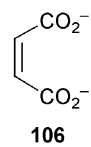
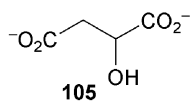
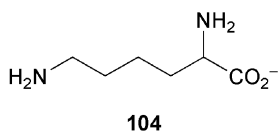
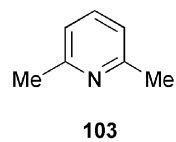
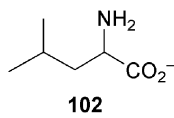
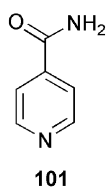
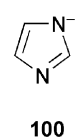
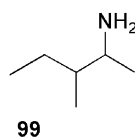
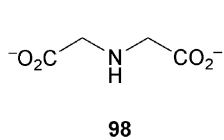
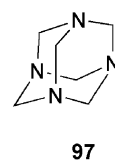
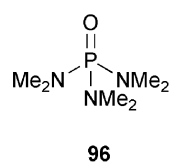
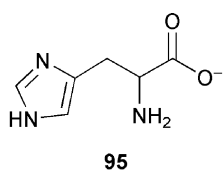
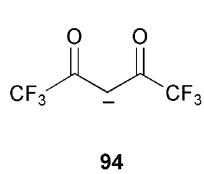
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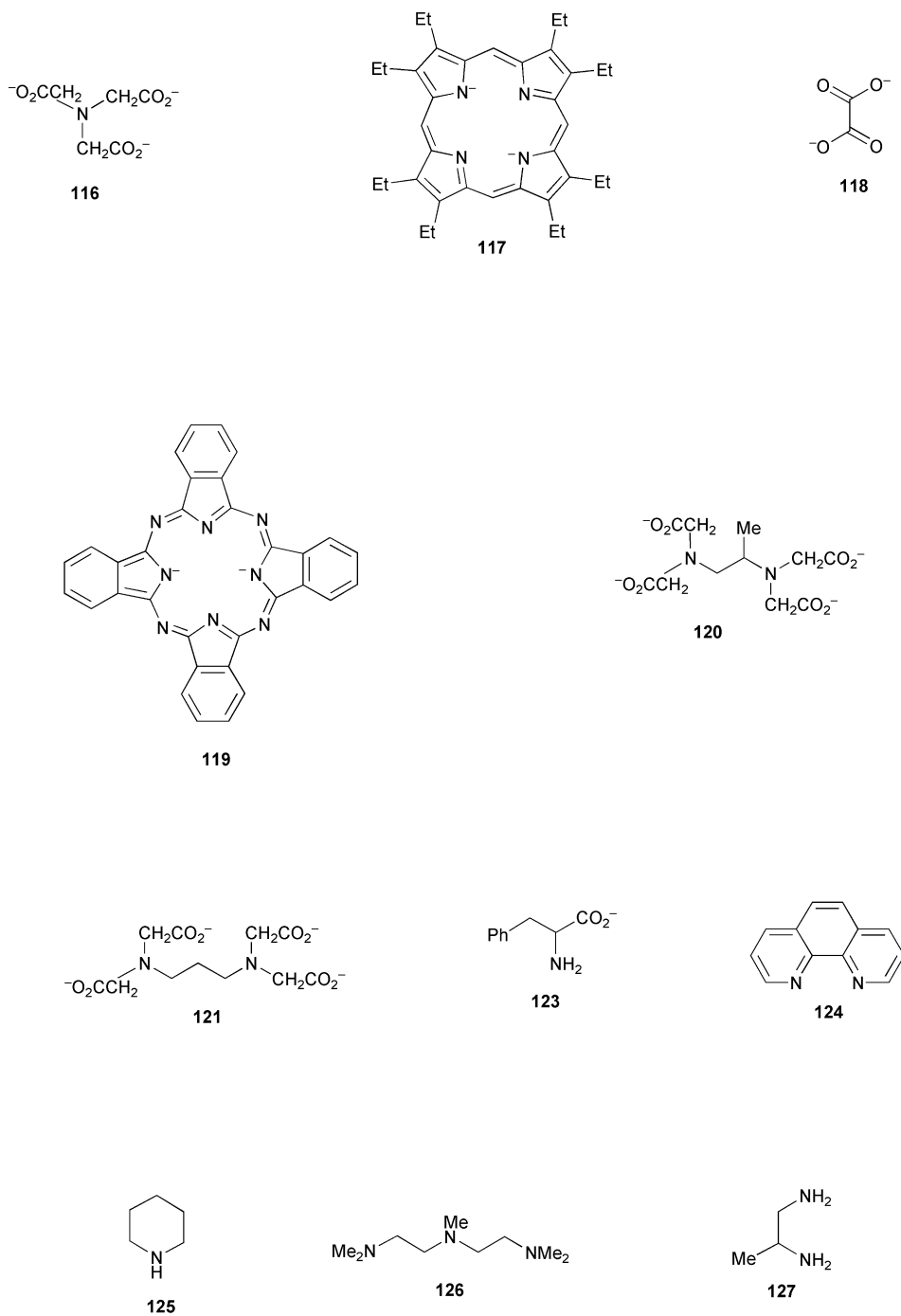
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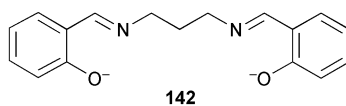
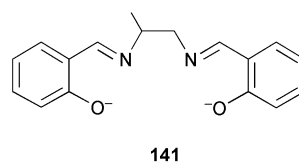
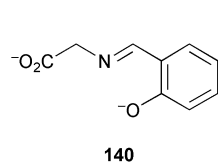
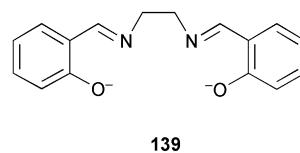
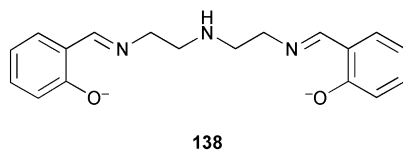
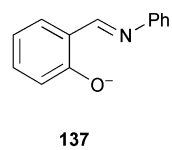
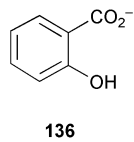
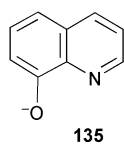
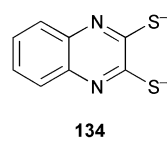
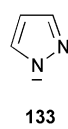
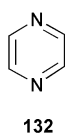
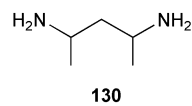
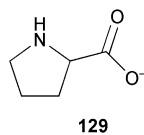
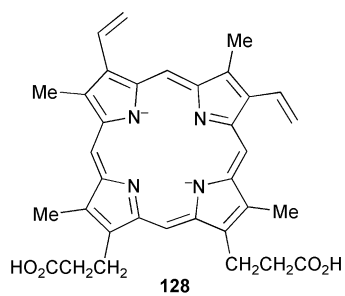
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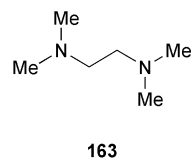
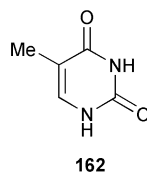
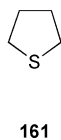
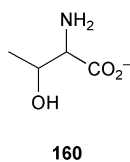
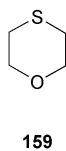
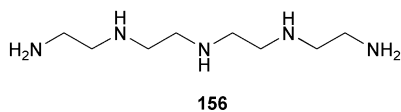
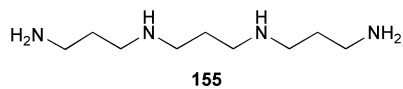
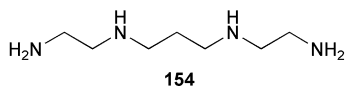
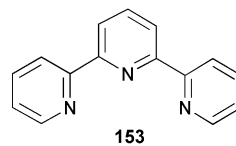
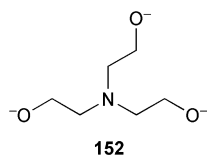
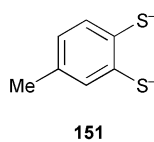
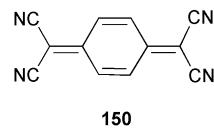
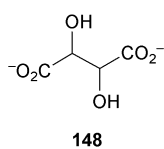
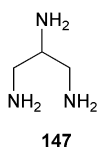
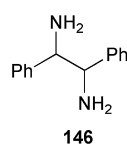
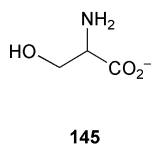
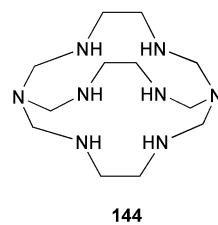
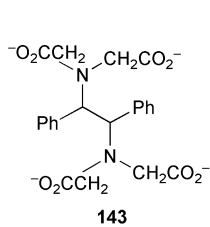
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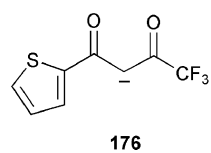
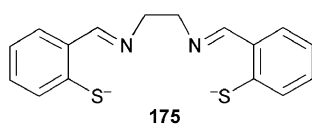
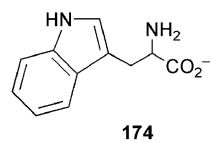
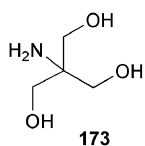
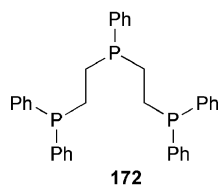
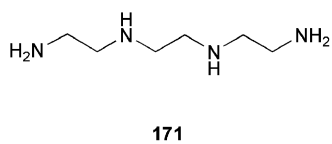
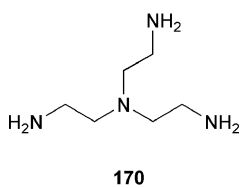
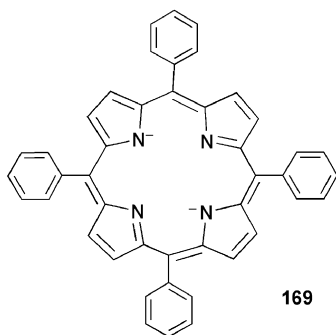
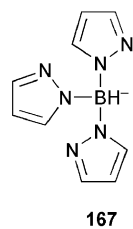
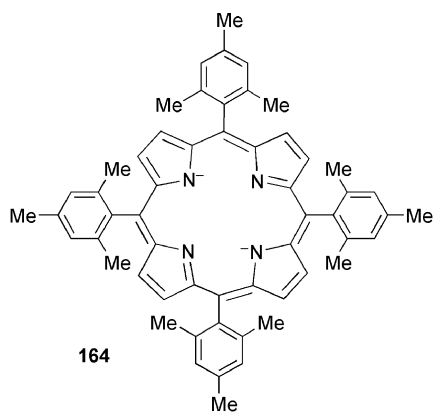
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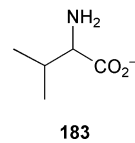
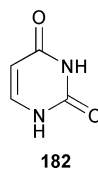
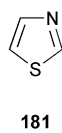
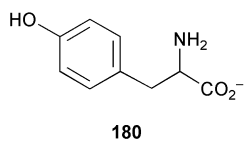
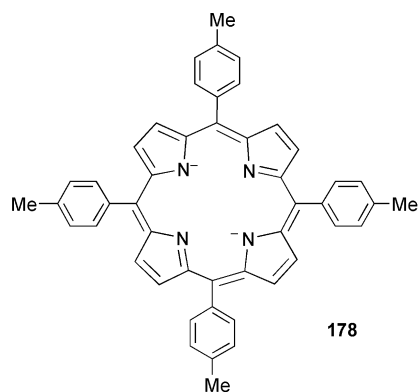
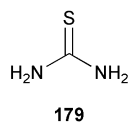
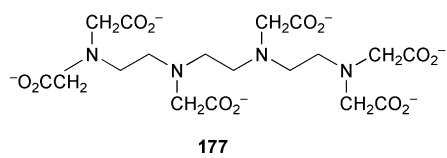
Table VIII *Continued*

Table IX Names of homoatomic, binary and certain other simple molecules, ions, compounds, radicals and substituent groups

This Table names a large number of homoatomic and binary compounds and species, and some heteropolynuclear entities, and thus may be used as a reference for names of simple compounds and a source of examples to guide in the naming of further compounds. It may be necessary to browse the Table to find (families of) compounds that match those of interest. For example, all the oxides of potassium are named; corresponding compounds of the other alkali metals, not included here, are named analogously. Several silicon and germanium hydride species are named; names for corresponding tin and lead species are not necessarily included.

Some inorganic acids and their corresponding bases are included in this Table, but more acid names are given in Tables IR-8.1 and IR-8.2. Only a few simple carbon-containing compounds and substituent groups are included. In particular, organic ligands belonging to the general classes alcohols, thiolates, phenolates, carboxylates, amines, phosphanes and arsanes as well as (partially) dehydronated amines, phosphanes and arsanes are generally *not* included. Their naming is described and exemplified in Section IR-9.2.2.3.

Entries in the first column are ordered alphabetically according to the formulae as they appear here. Formulae for *binary* species are written in this column according to the position of the two elements in Table VI (Section IR-4.4), *e.g.* ammonia is found under 'NH₃', but selane under 'H₂Se' and AlLi under 'LiAl'. In case of doubt, cross references should aid in finding the correct entry in the Table. However, in the first column, formulae of *ternary* and *quaternary* compounds are written strictly alphabetically, *e.g.* 'ClSCN^{•-}' is found under the entry 'CClNS', and 'HPO₄⁻' under 'HO₄P', and these formulae are ordered as described in Section IR-4.4.2.2. In the columns to the right of the first column, special formats may be used for formulae in order to stress a particular structure, *e.g.* under the entry 'BrHO₃' one finds 'HOBrO₂' rather than 'HBrO₃' or '[BrO₂(OH)]', the two formats presented in Table IR-8.2.

The symbol '⊂' is used for dividing names when this is made necessary by a line break. When the name is reconstructed from the name given in the Table, this symbol should be omitted. Thus, all *hyphens* in the Table are true parts of the names. The symbols '>' and '<' placed next to an element symbol both denote two single bonds connecting the atom in question to two other atoms.

For a given compound, the various systematic names, if applicable, are given in the order: stoichiometric names (Chapter IR-5), substitutive names (Chapter IR-6), additive names (Chapter IR-7) and hydrogen names (Section IR-8.4). Acceptable names which are not entirely systematic (or not formed according to any of the systems mentioned above) are given at the end after a semicolon. No order of preference is implied by the order in which formulae and names are listed, but in practice it may be useful to select particular formulae and names for particular uses. Thus, for sodium chloride the formula [NaCl] and the additive name 'chloridosodium' may be used specifically for the molecular compound, which can be regarded as a coordination compound, whereas 'sodium chloride' may be used, and traditionally *is* used, for the compound in general and for the solid with the composition NaCl. Corresponding remarks apply to a number of hydrides for which a stoichiometric name may be applied to the compound with the stoichiometry in question without further structural implications (such as 'aluminium trihydride' for AlH₃ or 'dihydrogen disulfide' for H₂S₂), whereas a parent hydride name (needed anyway for naming certain derivatives) or an additive name may be used to denote specifically the molecular compound or entity (such as 'alumane' or 'trihydroaluminium' for the molecular entity [AlH₃]) and 'disulfane' for HSSH).

Note from the examples above that in order to stress the distinctions discussed, the square brackets are sometimes used in the Table to enclose formulae for molecular entities that are otherwise often written with no enclosing marks. When a formula with square brackets is shown, there will also be a coordination-type additive name.

Formula for uncharged atom or group	Name			
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
Ac	actinium	actinium	actinide ^d	actinido
Ag	silver	silver	argentide	argentido
Al	aluminium	aluminium (general) Al ⁺ , aluminium(1+) Al ³⁺ , aluminium(3+)	aluminide (general) Al ⁻ , aluminide(1-)	aluminido (general) Al ⁻ , aluminido(1-)
AlCl	AlCl, aluminium monochloride [AlCl], chloridoaluminium	AlCl ⁺ , chloridoaluminium(1+)		
AlCl ₃ (see also Al ₂ Cl ₆)	AlCl ₃ , aluminium trichloride [AlCl ₃], trichloroalumane, trichloridoaluminium			
AlCl ₄			AlCl ₄ ⁻ , tetrachloroalumanuide, tetrachloridoaluminate(1-)	AlCl ₄ ⁻ , tetrachloroalumanuido, tetrachloridoaluminate(1-)
AlH	AlH, aluminium monohydride [AlH], λ ¹ -alumane (parent hydride name), hydridoaluminium	AlH ⁺ , hydridoaluminium(1+)		
AlH ₂	-AlH ₂ , alumanyl			
AlH ₃	AlH ₃ , aluminium trihydride [AlH ₃], alumane (parent hydride name), trihydridoaluminium	AlH ₃ ^{•+} , alumaniumyl, trihydridoaluminium(•1+)	AlH ₃ ^{•-} , alumanidyl, trihydridoaluminate(•1-) ^e	
AlH ₄			AlH ₄ ⁻ , alumanuide, tetrahydridoaluminate(1-)	AlH ₄ ⁻ , alumanuido, tetrahydridoaluminate(1-)
AlO	AlO, aluminium mon(o)oxide [AlO], oxidoaluminium	AlO ⁺ , oxidoaluminium(1+)	AlO ⁻ , oxidoaluminate(1-)	
AlSi	AlSi, aluminium monosilicide [AlSi], silicidoaluminium			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
Al ₂	Al ₂ , dialuminium		Al ₂ ⁻ , dialuminide(1-)	
Al ₂ Cl ₆	[Cl ₂ Al(μ-Cl) ₂ AlCl ₂], di-μ-chlorido-bis(dichlorido- O - aluminium)			
Al ₄			Al ₄ ²⁻ , tetraaluminide(2-)	
Am	americium	americium	americide	americido
Ar	argon	argon (general) Ar ⁺ , argon(1+)	argonide	argonido
ArBe		ArBe ⁺ , beryllidoargon(1+)		
ArF	ArF, argon monofluoride [ArF], fluoroargon	ArF ⁺ , fluoroargon(1+)		
ArHe		ArHe ⁺ , helidoargon(1+)		
ArLi		ArLi ⁺ , lithidoargon(1+)		
Ar ₂	Ar ₂ , diargon	Ar ₂ ⁺ , diargon(1+)		
As	arsenic >As-, arsanytriyl	arsenic	arsenide (general) As ³⁻ , arsenide(3-), arsanytriide; arsenide	arsenido (general) As ³⁻ , arsanytriido; arsenido
AsH	AsH, arsenic monohydride AsH ^{2•} , arsanylidene, hydridoarsenic(2•) >AsH, arsanydiyl =AsH, arsanylidene	AsH ²⁺ , arsanebis(ylum), hydridoarsenic(1+)	AsH ²⁻ , arsanydiide, hydridoarsenate(2-)	AsH ²⁻ , arsanydiido, hydridoarsenate(2-)

AsHO	>AsH(O), oxo- λ^5 -arsanedyl; arsonoyl = AsH(O), oxo- λ^5 -arsanylidene; arsonoylidene			
AsHO ₂	>AsO(OH), hydroxy(oxo)- λ^5 -arsanedyl; hydroxyarsoryl = AsO(OH), hydroxy(oxo)- λ^5 - arsanylidene; hydroxyarsorylidene	AsHO ₂ ²⁻ , hydridodioxidoarsenate(2-); arsonite	AsHO ₂ ²⁻ , hydridodioxidoarsenate(2-); arsonite	AsHO ₂ ²⁻ , hydridodioxidoarsenate(2-); arsonito
AsHO ₃		AsHO ₃ ²⁻ , hydridotrioxidoarsenate(2-); arsonate	AsHO ₃ ²⁻ , hydridotrioxidoarsenate(2-); arsonate	AsHO ₃ ²⁻ , hydridotrioxidoarsenate(2-); arsonato
AsH ₂	AsH ₂ , arsenic dihydride AsH ₂ [•] , arsanyl, dihydridoarsenic(•) -AsH ₂ , arsanyl	AsH ₂ ⁺ , arsanylium, dihydridoarsenic(1+)	AsH ₂ ⁺ , arsanylium, dihydridoarsenic(1+)	AsH ₂ ⁺ , arsanylium, dihydridoarsenic(1+)
AsH ₂ O	-AsH ₂ O, oxo- λ^5 -arsanyl; arsinoyl	AsH ₂ O ⁻ , dihydridoarsenate(1-); arsinite	AsH ₂ O ⁻ , dihydridoarsenate(1-); arsinite	AsH ₂ O ⁻ , dihydridoarsenate(1-); arsinito
AsH ₂ O ₂		AsH ₂ O ₂ ⁻ , dihydridodioxidoarsenate(1-); arsinate	AsH ₂ O ₂ ⁻ , dihydridodioxidoarsenate(1-); arsinate	AsH ₂ O ₂ ⁻ , dihydridodioxidoarsenate(1-); arsinato
AsH ₂ O ₃	-As(O)(OH) ₂ , dihydroxyoxo- λ^5 - arsanyl; dihydroxyarsoryl, arsono	AsO(OH) ₂ ⁻ , dihydroxydioxidoarsenate(1-)	AsO(OH) ₂ ⁻ , dihydroxydioxidoarsenate(1-)	AsO(OH) ₂ ⁻ , dihydroxydioxidoarsenate(1-)
AsH ₃	AsH ₃ , arsenic trihydride [AsH ₃], arsane (parent hydride name), trihydridoarsenic	AsH ₃ ^{•+} , arsaniumyl, trihydridoarsenic(•1+) -AsH ₃ ⁺ , arsaniumyl	AsH ₃ ^{•+} , arsaniumyl, trihydridoarsenate(•1+) ^e	
AsH ₄	-AsH ₄ , λ^5 -arsanyl	AsH ₄ ⁺ , arsanium, tetrahydridoarsenic(1+)		
AsH ₅	AsH ₅ , arsenic pentahydride [AsH ₅], λ^5 -arsane (parent hydride name), pentahydridoarsenic			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
AsO	>As(O)–, oxo- λ^5 -arsanetriyl; arsoryl =As(O)–, oxo- λ^5 -arsanylidene; arsorylidene $\equiv$ As(O), oxo- λ^5 -arsanyldiyl; arsoryldiyl			
AsO ₃			AsO ₃ ^{3–} , trioxidoarsenate(3–); arsenite, arsorite –As(=O)(O [–]) ₂ , dioxidooxo- λ^5 -arsanyl; arsonato	AsO ₃ ^{3–} , trioxidoarsenato(3–); arsenito, arsorito
AsO ₄			AsO ₄ ^{3–} , tetraoxidoarsenate(3–); arsenate, arsorate	AsO ₄ ^{3–} , tetraoxidoarsenato(3–); arsenato, arsorato
AsS ₄			AsS ₄ ^{3–} , tetrasulfidoarsenate(3–)	AsS ₄ ^{3–} , tetrasulfidoarsenato(3–)
As ₂ H			HAs=As [–] , diarsenide HAsAs ^{3–} , diarsanetriide	HAs=As [–] , diarsenido HAsAs ^{3–} , diarsanetriido
As ₂ H ₂	HAs=AsH, diarsene		H ₂ AsAs ^{2–} , diarsane-1,1-diide HAsAsH ^{2–} , diarsane-1,2-diide	HAs=AsH, diarsene H ₂ AsAs ^{2–} , diarsane-1,1-diido HAsAsH ^{2–} , diarsane-1,2-diido
As ₂ H ₄	H ₂ AsAsH ₂ , diarsane			H ₂ AsAsH ₂ , diarsane
As ₄	As ₄ , tetraarsenic			As ₄ , tetraarsenic
At	astatine (general) At [•] , astatine(•), monoastatine	astatine	At [–] , astatide(1–); astatide	astatido (general) At [–] , astatido(1–); astatido
AtH, see HAt				

At ₂	At ₂ , diastatine		auride		aurido
Au	gold	gold (general) Au ⁺ , gold(1+) Au ³⁺ , gold(3+)			aurido
B	boron >B-, boranetriyl ≡B, boranylidene	boron (general) B ⁺ , boron(1+) B ³⁺ , boron(3+)	boride (general) B ⁻ , boride(1-) B ³⁻ , boride(3-); boride		borido (general) B ⁻ , borido(1-) B ³⁻ , borido(3-); borido
BH	>BH, boranediyl =BH, boranylidene	BH ²⁺ , boranebis(ylium), hydridoboron(2+)	BH ²⁻ , boranediide, hydridoborate(2-)		BH ²⁻ , boranediido, hydridoborate(2-)
BHO ₃			BO ₂ (OH) ²⁻ , hydroxidodioxidoborate(2-); hydrogenborate		BO ₂ (OH) ²⁻ , hydroxidodioxidoborate(2-); hydrogenborato
BH ₂	-BH ₂ , boranyl	BH ₂ ⁺ , boranylium, dihydridoboron(1+)	BH ₂ ⁻ , boranide, dihydridoborate(2-)		BH ₂ ⁻ , boranido, dihydridoborate(2-)
BH ₂ O	-BH(OH), hydroxyboranyl				
BH ₂ O ₂	-B(OH) ₂ , dihydroxyboranyl; borono				
BH ₃	BH ₃ , boron trihydride [BH ₃], borane (parent hydride name), trihydridoboron	BH ₃ ^{•+} , boraniumyl, trihydridoboron(•1+)	BH ₃ ^{•-} boranuidyl, trihydridoborate(•1-) ^e -BH ₃ ⁻ , boranuidyl		BH ₃ ^{•-} , trihydridoborato(•1-)
BH ₄		BH ₄ ⁺ , boranium, tetrahydridoboron(1+)	BH ₄ ⁻ , boranuide, tetrahydridoborate(1-)		BH ₄ ⁻ , boranuideo, tetrahydridoborato(1-)
BO	BO, boron mon(o)oxide [BO], oxidoboron	BO ⁺ , oxidoboron(1+)	BO ⁻ , oxidoborate(1-)		BO ⁻ , oxidoborato(1-)
BO ₂			(BO ₂ ⁻) _n = (OBO) _n ⁿ⁻ , <i>catena</i> -poly[(oxidoborate-μ- oxido)(1-)]; metaborate		
BO ₃			BO ₃ ³⁻ , trioxidoborate(3-); borate		BO ₃ ³⁻ , trioxidoborato(3-); borato
Ba	barium		baride		barido
BaO	barium oxide	barium			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
BaO ₂	Ba ²⁺ O ₂ ²⁻ , barium dioxide(2-); barium peroxide			
Be	beryllium	beryllium (general) Be ⁺ , beryllium(1+) Be ²⁺ , beryllium(2+)	beryllide	beryllido
BeH	BeH, beryllium monohydride [BeH], hydridoberyllium	BeH ⁺ , hydridoberyllium(1+)	BeH ⁻ , hydridoberyllate(1-)	BeH ⁻ , hydridoberyllato(1-)
Bh	bohrium	bohrium	bohride	bohrido
Bi	bismuth	bismuth	bismuthide (general) Bi ³⁻ , bismuthide(3-), bismuthanetriide; bismuthide	bismuthido (general) Bi ³⁻ , bismuthido(3-), bismuthanetriido; bismuthido
BiH	>BiH, bismuthanediyl =BiH, bismuthanylidene BiH ^{2•} , bismuthanylidene, hydridobismuth(2•)	BiH ²⁺ , bismuthanebis(ylum), hydridobismuth(2+)	BiH ²⁻ , bismuthanediide, hydridobismuthate(2-)	BiH ²⁻ , bismuthanediido, hydridobismuthato(2-)
BiH ₂	-BiH ₂ , bismuthanyl BiH ₂ [•] , bismuthanyl, dihydridobismuth(•)	BiH ₂ ⁺ , bismuthanylium, dihydridobismuth(1+)	BiH ₂ ⁻ , bismuthanide, dihydridobismuthate(1-)	BiH ₂ ⁻ , bismuthanido, dihydridobismuthato(1-)
BiH ₃	BiH ₃ , bismuth trihydride [BiH ₃], bismuthane (parent hydride name), trihydridobismuth =BiH ₃ , λ ⁵ -bismuthanylidene	BiH ₃ ^{•+} , bismuthaniumyl, trihydridobismuth(•1+)	BiH ₃ ^{•-} , bismuthanuidyl, trihydridobismuthate(•1-) ^e	

BiH ₄		BiH ₄ ⁺ , bismuthanium, tetrahydridobismuth(1+)		
Bi ₅		Bi ₅ ⁴⁺ , pentabismuth(4+)		
Bk	berkelium	berkelium	berkelide	berkelido
Br	bromine (general) Br [•] , bromine(•), monobromine -Br, bromo	bromine (general) Br ⁺ , bromine(1+)	bromide (general) Br ⁻ , bromide(1-); bromide	bromido (general) Br ⁻ , bromido(1-); bromido
BrCN	BrCN, cyanobromane, bromidinitridocarbon			
BrH, see HBr				
BrHO	HOBr, bromanol, hydrodibromine ^f ; hypobromous acid			
BrHO ₂	HOBrO, hydroxy-λ ³ -bromanone, hydrodioxidobromine; bromous acid			
BrHO ₃	HOBrO ₂ , hydroxy-λ ⁵ -bromanedi-one, hydrodioxidobromine; bromic acid			
BrHO ₄	HOBrO ₃ , hydroxy-λ ⁷ -bromanetrione, hydrodioxidobromine; perbromic acid			
Br ₂	Br ₂ , dibromine	Br ₂ ^{•+} , dibromine(•1+)	Br ₂ ^{•-} , dibromide(•1-)	Br ₂ , dibromine
Br ₃	Br ₃ , tribromine		Br ₃ ⁻ , tribromide(1-); tribromide	Br ₃ ⁻ , tribromido(1-); tribromido
C	carbon (general) C, monocarbon >C<, methanetetrayl =C=, methanediylidene	carbon (general) C ⁺ , carbon(1+)	carbide (general) C ⁻ , carbide(1-) C ⁴⁻ , carbide(4-), methanetetraide; carbide	carbido (general) C ⁻ , carbido(1-) C ⁴⁻ , carbido(4-), methanetetrayl, methanetetraido
CClNS			ClSCN ^{•-} , (chloridosulfato)nitrido⇌ carbonate(•1-)	

Table IX Continued

Formula for uncharged atom or group	Name			
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
CH	CH [•] , hydridocarbon(•) CH ^{3•} , methylidyne, hydridocarbon(3•), carbyne ≡CH, methyldyne -CH=, methanylylidene -CH<, methanetriyl	CH ⁺ , λ ² -methanylium, hydridocarbon(1+)	CH ⁻ , λ ² -methanide, hydridocarbonate(1-) CH ³⁻ , methanetriide, hydridocarbonate(3-)	CH ⁻ , λ ² -methanido, hydridocarbonato(1-) CH ³⁻ , methanetriyl, methanetriido, hydridocarbonato(3-)
CHN	HCN, hydrogen cyanide HCN = [CHN], methanenitrile, hydridonitridocarbon; formonitrile >C=NH, carbonimidoyl =C=NH, iminomethylidene, carbonimidoylidene			
CHNO	HCNO = [N(CH)O], formonitrile oxide, (hydridocarbonato)oxidonitrogen HNCO = [C(NH)O], (hydridonitrato)oxidocarbon; isocyanic acid HOCN = [C(OH)N], hydroxidonitridocarbon; cyanic acid HONC = [NC(OH)], λ ² -methylidenehydroxylamine, carbidohydroxidonitrogen		HNCO ^{•-} , (hydridonitrato)oxido⊖ carbonato(•1-) HOCN ^{•-} , hydroxidonitridocarbonato(•1-)	HNCO ^{•-} , (hydridonitrato)oxido⊖ carbonato(•1-) HOCN ^{•-} , hydroxidonitridocarbonato(•1-)

CHNOS			$\text{HONCS}^{\bullet-}$, (hydroxidonitrato)sulfido $\oslash$ carbonato($\bullet 1-$) $\text{HOSCN}^{\bullet-}$, (hydroxidosulfato)nitrido $\oslash$ carbonato($\bullet 1-$)	$\text{HONCS}^{\bullet-}$, (hydroxidonitrato)sulfido $\oslash$ carbonato($\bullet 1-$) $\text{HOSCN}^{\bullet-}$, (hydroxidosulfato)nitrido $\oslash$ carbonato($\bullet 1-$)
CHNO ₂			$\text{HOOCN}^{\bullet-}$, (dioxidanido)nitridocarbonate($\bullet 1-$) $\text{HONCO}^{\bullet-}$, (hydroxidonitrato)oxido $\oslash$ carbonato($\bullet 1-$)	$\text{HOOCN}^{\bullet-}$, (dioxidanido)nitrido $\oslash$ carbonato($\bullet 1-$) $\text{HONCO}^{\bullet-}$, (hydroxidonitrato)oxido $\oslash$ carbonato($\bullet 1-$)
CHNS	$\text{HCNS} = \text{HC}\equiv\text{N}^+\text{S}^-$ $= [\text{N}(\text{CH})\text{S}]$, (methyldyneammonium) $\oslash$ sulfanide, (hydridocarbonato)sulfidonitrogen $\text{HNCS} = [\text{C}(\text{NH})\text{S}]$, (hydridonitrato)sulfidocarbon; isothiocyanic acid $\text{HSCN} = [\text{CN}(\text{SH})]$, nitridosulfanidocarbon; thiocyanic acid $\text{HSNC} = [\text{NC}(\text{SH})]$, λ^2 -methylenethiohydroxylamine, carbidosulfanidonitrogen			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
CHNSe	HCNSe = $\text{HC}\equiv\text{N}^+\text{Se}^-$ = [N(CH)Se], (methyldiyneammonium)⊖ selenide, (hydridocarbonato)selenidonitrogen HNCS _e = [C(NH)Se], (hydridonitrato)selenidocarbon; isoselenocyanic acid HSeCN = [CN(SeH)], nitridoselanidocarbon; selenocyanic acid HSeNC = [NC(SeH)], λ ² -methylideneseleno⊖ hydroxylamine, carbidoselanidonitrogen			
CHO	HCO [•] , oxomethyl, hydriooxidocarbon(•) -CH(O), methanoyl, formyl			
CHOS ₂	HOCS ₂ [•] , hydroxidodisulfidocarbon(•)			
CHO ₂	HOCO [•] , hydroxidooxidocarbon(•)			

CHO ₃	HOCO ₂ [•] , hydroxidodioxidocarbon(•) HOOCO [•] , (dioxidanido)oxidocarbon(•)		HCO ₃ ⁻ , hydroxidodioxidocarbonate(1-); hydrogencarbonate	HCO ₃ ⁻ , hydroxidodioxidocarbonato(1-); hydrogencarbonato
CH ₂	CH ₂ , λ ² -methane CH ₂ ^{2•} , methylenide, dihydridocarbon(2•); carbene >CH ₂ , methanediyl, methylene =CH ₂ , methylenide		CH ₂ ²⁻ , methanediide, dihydridocarbonate(2-) -CH ₂ ⁻ , methanidyl	>CH ₂ , methanediyl, methylene =CH ₂ , methylenide CH ₂ ²⁻ , methanediido, dihydridocarbonato(2-)
CH ₂ N	H ₂ CN [•] , dihydridonitridocarbon(•)			
CH ₂ NO	H ₂ NCO [•] , (dihydridonitrato)oxidocarbon(•) HNCOH [•] , (hydridonitrato)hydroxido carbon(•)			
CH ₃	CH ₃ [•] , methyl -CH ₃ <i>or</i> -Me, methyl	CH ₃ ⁺ , methylum, trihydridocarbon(1+)	CH ₃ ⁻ , methanide, trihydridocarbonate(1-)	CH ₃ ⁻ , methyl, methanido, trihydridocarbonato(1-)
CH ₄	CH ₄ , methane (parent hydride name), tetrahydridocarbon	CH ₄ ^{•+} , methaniumyl, tetrahydridocarbon(•1+)	CH ₄ ^{•-} , methanuidyl, tetrahydridocarbonate(•1-) ^e	
CH ₅		CH ₅ ⁺ , methanium, pentahydridocarbon(1+)		
CN	CN [•] , nitridocarbon(•); cyanyl -CN, cyano -NC, isocyano	CN ⁺ , azanylidynemethylum, nitridocarbon(1+)	CN ⁻ , nitridocarbonate(1-); cyanide	nitridocarbonato (general) CN ⁻ , nitridocarbonato(1-); cyanido = [nitridocarbonato(1-)-κC]
CN ₂			NCN ²⁻ , dinitridocarbonate(2-)	NCN ²⁻ , dinitridocarbonato(2-)
CNO	OCN [•] , nitridooxidocarbon(•) -OCN, cyanato -NCO, isocyanato -ONC, λ ² -methylideneazanylideneoxy -CNO, (oxo-λ ² -azanylidyne)methyl		OCN ⁻ , nitridooxidocarbonate(1-); cyanate ONC ⁻ , carbidooxidonitrate(1-); fulminate OCN ^{•2-} , nitridooxidocarbonate(•2-)	OCN ⁻ , nitridooxidocarbonato(1-); cyanato ONC ⁻ , carbidooxidonitrato(1-); fulminato

Table IX Continued

Formula for uncharged atom or group	Name			
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
CNS	SCN [•] , nitridosulfidocarbon(•) –SCN, thiocyanato –NCS, isothiocyanato –SNC, λ^2 -methylidene- $\text{C}\equiv$ azanylylidenesulfanediyyl –CNS, (sulfanylidene- λ^5 - azanylidene)methyl		SCN [–] , nitridosulfidocarbonato(1–); thiocyanato SNC [–] , carbidosulfidonitrato(1–)	SCN [–] , nitridosulfidocarbonato(1–); thiocyanato SNC [–] , carbidosulfidonitrato(1–)
CNSe	SeCN [•] , nitridoselenidocarbon(•) –SeCN, selenocyanato –NCSe, isoselenocyanato –SeNC, λ^2 -methylidene- $\text{C}\equiv$ azanylylideneselenanediyyl –CNSe, (selanylidene- λ^5 - azanylidene)methyl		SeCN [–] , nitridoselenidocarbonato(1–); selenocyanato SeNC [–] , carbidoselenidonitrato(1–)	SeCN [–] , nitridoselenidocarbonato(1–); selenocyanato SeNC [–] , carbidoselenidonitrato(1–)
CO	CO, carbon mon(o)oxide >C=O, carbonyl =C=O, carbonylidene	CO ^{•+} , oxidocarbon(•1+) CO ²⁺ , oxidocarbon(2+)	CO ^{•–} , oxidocarbonato(•1–)	CO, oxidocarbon, oxidocarbonato (general); carbonyl = oxidocarbon- κC (general) CO ^{•+} , oxidocarbon(•1+) CO ^{•–} , oxidocarbonato(•1–)
COS	C(O)S, carbonyl sulfide, oxidosulfidocarbon			
CO ₂	CO ₂ , carbon dioxide, dioxidocarbon		CO ₂ ^{•–} , oxidooxomethyl, dioxidocarbonato(•1–)	CO ₂ , dioxidocarbon CO ₂ ^{•–} , oxidooxomethyl, dioxidocarbonato(•1–)

CO_3			$\text{CO}_3^{\bullet-}$, trioxidocarbonate($\bullet 1-$), $\text{OCO}^{\bullet-}$, (dioxido)oxidocarbonate($\bullet 1-$), oxidoperoxidocarbonate($\bullet 1-$) CO_3^{2-} , trioxidocarbonate($2-$); carbonate	CO_3^{2-} , trioxidocarbonato($2-$); carbonato
CS	carbon monosulfide >C=S, carbonothioyl; thiocarbonyl =C=S, carbonothioylidene	$\text{CS}^{\bullet+}$, sulfidocarbon($\bullet 1+$)	$\text{CS}^{\bullet-}$, sulfidocarbonate($\bullet 1-$)	CS, sulfidocarbon, sulfidocarbonato, thiocarbonyl (general); $\text{CS}^{\bullet+}$, sulfidocarbon($\bullet 1+$) $\text{CS}^{\bullet-}$, sulfidocarbonato($\bullet 1-$)
CS_2	CS_2 , disulfidocarbon, carbon disulfide		$\text{CS}_2^{\bullet-}$, sulfidothioxomethyl, disulfidocarbonate($\bullet 1-$)	CS_2 , disulfidocarbon $\text{CS}_2^{\bullet-}$, sulfidothioxomethyl, disulfidocarbonato($\bullet 1-$)
CS_3			CS_3^{2-} , trisulfidocarbonate($2-$)	CS_3^{2-} , trisulfidocarbonato($2-$)
C_2	C_2 , dicarbon	C_2^+ , dicarbon($1+$)	C_2^- , dicarbide($1-$) C_2^{2-} , dicarbide($2-$), ethynediide, acetylenediide; acetylde	dicarbido (general) C_2^{2-} , dicarbido($2-$), ethynediido, ethyne-1,2-diyl
C_2H	$\text{HCC}^{\bullet}$, ethynyl, hydridodicarbon($\bullet$)			
C_2N_2	NCCN , ethanedinitrile, bis(nitridocarbon)($C-C$); oxalonitrile		$\text{NCCN}^{\bullet-}$, bis(nitridocarbonato)($C-C$)($\bullet 1-$)	
$\text{C}_2\text{N}_2\text{O}_2$	NCOOCN , dioxidanedicarbonitrile, bis[cyanidooxygen]($O-O$)		$\text{NCOOCN}^{\bullet-}$, bis[cyanidooxygenate]($O-O$)($\bullet 1-$) ^e $\text{OCNNCO}^{\bullet-}$, bis(carbonylnitrate)($N-N$)($\bullet 1-$) ^e	$\text{NCOOCN}^{\bullet-}$, bis[cyanidooxygenate]($O-O$)($\bullet 1-$) $\text{OCNNCO}^{\bullet-}$, bis(carbonylnitrate)($N-N$)($\bullet 1-$)
$\text{C}_2\text{N}_2\text{S}_2$	NCSSCN , disulfanedicarbonitrile, bis[cyanidosulfur]($S-S$)		$\text{NCSSCN}^{\bullet-}$, bis[cyanidosulfate]($S-S$)($\bullet 1-$) ^e	$\text{NCSSCN}^{\bullet-}$, bis[cyanidosulfato]($S-S$)($\bullet 1-$)
C_3O_2	C_3O_2 , tricarbon dioxide $\text{O}=\text{C}=\text{C}=\text{C}=\text{O}$, propa-1,2-diene-1,3-dione			
C_{12}O_9	C_{12}O_9 , dodecacarbon nonaoxide			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
Ca	calcium	calcium (general) Ca ²⁺ , calcium(2+)	calcide	calcido
Cd	cadmium	cadmium (general) Cd ²⁺ , cadmium(2+)	cadmide	cadmido
Ce	cerium	cerium (general) Ce ³⁺ , cerium(3+) Ce ⁴⁺ , cerium(4+)	ceride	cerido
Cf	californium	californium	californide	californido
Cl	chlorine (general) Cl [•] , chlorine(•), monochlorine -Cl, chloro	chlorine (general) Cl ⁺ , chlorine(1+)	chloride (general) Cl ⁻ , chloride(1-); chloride	chlorido (general) Cl ⁻ , chlorido(1-); chlorido
ClF	ClF, fluoridochlorine, chlorine monofluoride	ClF ⁺ , fluoridochlorine(1+)		
ClF ₂			ClF ₂ ⁻ , difluoridochlorate(1-)	ClF ₂ ⁻ , difluoridochlorato(1-)
ClF ₄		ClF ₄ ⁺ , tetrafluoridochlorine(1+)	ClF ₄ ⁻ , tetrafluoridochlorate(1-)	ClF ₄ ⁻ , tetrafluoridochlorato(1-)
ClH, see HCl				
ClHN			NHCl ⁻ , chloroazanide, chloridohydridonitrate(1-)	NHCl ⁻ , chloroazanido, chloridohydridonitrato(1-)
ClHO	HOCl, chloranol, hydroxidochlorine ^f ; hypochlorous acid		HOCl ^{•-} , hydroxidochlorate(•1-)	
ClHO ₂	HOClO, hydroxy-λ ³ -chloranone, hydroxidooxidochlorine; chlorous acid			

ClHO ₃	HOClO ₂ , hydroxy-λ ⁵ -chloranediene, hydroxidodioxichlorine; chloric acid				
ClHO ₄	HOClO ₃ , hydroxy-λ ⁷ -chloranetrione, hydroxidotrioxichlorine; perchloric acid				
Cl ₂	Cl ₂ , dichlorine	Cl ₂ ^{•+} , dichlorine(•1+)	Cl ₂ ^{•-} , dichloride(•1-)	Cl ₂ , dichlorine Cl ₂ ^{•-} , dichlorido(•1-)	
Cl ₂ OP	-P(Cl ₂ (O), dichlorooxo-λ ⁵ -phosphanyl, phosphorodichloridoyl				
Cl ₄		Cl ₄ ⁺ , tetrachlorine(1+)			
Cm	curium	curium	curide	curido	
Co	cobalt	cobalt (general) Co ²⁺ , cobalt(2+) Co ³⁺ , cobalt(3+)	cobaltide	cobaltido	
Cr	chromium	chromium (general) Cr ²⁺ , chromium(2+) Cr ³⁺ , chromium(3+)	chromide	chromido	
CrO	CrO, chromium mon(o)oxide, chromium(II) oxide				
CrO ₂	CrO ₂ , chromium dioxide, chromium(IV) oxide				
CrO ₃	CrO ₃ , chromium trioxide, chromium(VI) oxide				
CrO ₄	[Cr(O ₂) ₂], dioxochromium		CrO ₄ ²⁻ , tetraoxidochromate(2-); chromate	CrO ₄ ²⁻ , tetraoxidochromato(2-); chromato	
			CrO ₄ ³⁻ , tetraoxidochromate(3-)	CrO ₄ ³⁻ , tetraoxidochromato(3-)	
			CrO ₄ ⁴⁻ , tetraoxidochromate(4-)	CrO ₄ ⁴⁻ , tetraoxidochromato(4-)	
CrO ₅	[CrO(O ₂) ₂], oxidodiperoxidochromium				

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
CrO ₆			CrO ₂ (O ₂) ₂ ²⁻ , dioxidodiperoxidochromate(2-)	
CrO ₈			Cr(O ₂) ₄ ²⁻ , tetraperoxidochromate(2-) Cr(O ₂) ₄ ³⁻ , tetraperoxidochromate(3-)	
Cr ₂ O ₃	Cr ₂ O ₃ , dichromium trioxide, chromium(III) oxide			
Cr ₂ O ₇			Cr ₂ O ₇ ²⁻ , heptaoxidodichromate(2-) O ₃ CrOCrO ₃ ²⁻ , μ -oxido-bis(trioxidochromate)(2-); dichromate	Cr ₂ O ₇ ²⁻ , heptaoxidodichromato(2-) O ₃ CrOCrO ₃ ²⁻ , μ -oxido-bis(trioxidochromato)(2-); dichromato
Cs	caesium	caesium	caeside	caesido
Cu	copper	copper (general) Cu ⁺ , copper(1+) Cu ²⁺ , copper(2+)	cupride	cuprido
D, see H				
D ₂ , see H ₂				
D ₂ O, see H ₂ O				
Db	dubnium	dubnium	dubnide	dubnido
Ds	darmstadtium	darmstadtium	darmstadtide	darmstadtido
Dy	dysprosium	dysprosium	dysproside	dysprosido
Er	erbium	erbium	erbide	erbido

Es	einsteinium	einsteinium	einsteinide	einsteinido	
Eu	euprium	euprium	eupride	euprido	
F	fluorine F [•] ; fluorine(•), monofluorine -F, fluoro	fluorine (general) F ⁺ , fluorine(1+)	fluoride (general) F ⁻ , fluoride(1-); fluoride	F ⁻ , fluorido(1-); fluorido	
FH, see HF					
FHO	HOF, fluoranol, fluoridohydridoxygen				
FNS	NSF, fluoridonitridosulfur				
FN ₃	FNNN, fluorido-1κF-trinitrogen(2 N-N)				
FO, see OF					
F ₂	F ₂ , difluorine	F ₂ ⁺ , difluorine(•1+)	F ₂ ⁻ , difluoride(•1-)	F ₂ , difluorine	
F ₂ N ₂	FN=NF, difluorido-1κF,2κF- dinitrogen(N-N), difluorodiazene				
Fe	iron	iron (general) Fe ²⁺ , iron(2+) Fe ³⁺ , iron(3+)	ferride	ferrido	
Fm	fermium	fermium	fermide	fermido	
Fr	francium	francium	francide	francido	
Ga	gallium	gallium	gallide	gallido	
GaH ₂	-GaH ₂ , gallanyl				
GaH ₃	GaH ₃ , gallium trihydride [GaH ₃], gallane (parent hydride name), trihydridogallium				
Gd	gadolinium	gadolinium	gadolinide	gadolinido	
Ge	germanium >Ge<, germanetetrayl =Ge=, germanediylidene >GeH-, germanetriyl =GeH-, germanylylidene ≡GeH, germlyidyne	germanium (general) Ge ²⁺ , germanium(2+) Ge ⁴⁺ , germanium(4+)	germide (general) Ge ⁴⁻ , germide(4-); germide	germido (general) Ge ⁴⁻ , germido(4-); germido	
GeH					

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
GeH ₂	>GeH ₂ , germandiyl =GeH ₂ , germalidene			
GeH ₃	–GeH ₃ , germyl	GeH ₃ ⁺ , germylum, trihydridogermanium(1+)	GeH ₃ [–] , germanide, trihydridogermanate(1–)	GeH ₃ [–] , germanido, trihydridogermanato(1–)
GeH ₄	GeH ₄ , germane (parent hydride name), tetrahydridogermanium			
Ge ₄				
H	hydrogen H [•] , hydrogen(•), monohydrogen (natural or unspecified isotopic composition) ¹ H [•] , protium(•), monoprotrium ² H [•] = D [•] , deuterium(•), monodeuterium ³ H [•] = T [•] , tritium(•), monotritium	hydrogen (general) H ⁺ , hydrogen(1+), hydron (natural or unspecified isotopic composition) ¹ H ⁺ , protium(1+), proton ² H ⁺ = D ⁺ , deuterium(1+), deuteron ³ H ⁺ = T ⁺ , tritium(1+), triton	Ge ₄ ^{4–} , tetragermide(4–) hydride (general) H [–] , hydride (natural or unspecified isotopic composition) ¹ H [–] , protide ² H [–] = D [–] , deuteride ³ H [–] = T [–] , tritide	hydrido protido deuterido tritido
HAt	HAt, hydrogen astatide [HAt], astatidohydrogen			
HBr	HBr, hydrogen bromide [HBr], bromane (parent hydride name), bromidohydrogen			
HCO, see CHO				

HCl	HCl, hydrogen chloride [HCl], chlorane (parent hydride name), chloridohydrogen	HCl ⁺ , chloraniumyl, chloridohydrogen(•1+)	
HF	HF, hydrogen fluoride [HF], fluorane (parent hydride name), fluoridohydrogen	HF ⁺ , fluoraniumyl, fluoridohydrogen(•1+)	
HF ₂			FHF ⁻ , fluoro fluoranuide, μ-hydrifluoridofluorane(1-), difluoridohydrogenate(1-)
HI	HI, hydrogen iodide [HI], iodane (parent hydride name), iodidohydrogen		
HIO	HOI, iodanol, hydroxidoiodine ¹ ; hypoiodous acid		
HIO ₂	HOIO, hydroxy-λ ³ -iodanone, hydroxidoiodine; iodous acid		
HIO ₃	HOIO ₂ , hydroxy-λ ⁵ -iodanedione, hydroxidoiodine; iodic acid		HOIO ₂ ^{•-} , hydroxidoiodine(•1-)
HIO ₄	HOIO ₃ , hydroxy-λ ⁷ -iodanetrione, hydroxidoiodine; periodic acid		
H _n N _m , see N _m H _n			
HMnO ₄	HMnO ₄ = [MnO ₃ (OH)], hydroxido trioxomanganese		HMnO ₄ ⁻ = [MnO ₃ (OH)] ⁻ , hydroxido trioxomanganate(1-)

Table IX Continued

Formula for uncharged atom or group	Name			
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
HNO	HNO = [NH(O)], azanone, hydridooidonitrogen HON ^{2•} , hydroxidonitrogen(2•) >NH(O), oxo-λ ⁵ -azanediyl; azonoyl =NH(O), oxo-λ ⁵ -azanylidene; azonoylidene >N-OH, hydroxyazanediyl =N-OH, hydroxyazanylidene; hydroxyimino	HNO ^{•+} = [NH(O)] ^{•+} , hydridooidonitrogen(•1+)	HON ²⁻ , hydroxidonitrate(2-)	HON ²⁻ , hydroxidonitrate(2-)
HNO ₂	HNO ₂ = [NO(OH)], hydrodooxidonitrogen; nitrous acid >N(O)(OH), hydroxyoxo-λ ⁵ -azanediyl; hydroxyazoryl =N(O)(OH), hydroxyoxo-λ ⁵ -azanylidene; hydroxyazorylidene			
HNO ₃	HNO ₃ = [NO ₂ (OH)], hydrodiodioxidonitrogen; nitric acid HNO(O ₂) = [NO(OOH)], dioxidanidooxidonitrogen, peroxynitrous acid			

HNO ₄	HNO ₄ = [NO ₂ (OOH)], (dioxidanido)dioxidonitrogen; peroxynitric acid			
HNS	>S(=NH), imino-λ ⁴ -sulfanediyl; sulfiminidoyl			
HN ₂ O ₂	–NHNO ₂ , nitrozazanyl, nitroamino		[HON=NO] [–] , 2-hydroxydiazen-1-olate, hydroxido-1κO-oxido-2κO- dinitrate(N–N)(1–)	
HN ₂ O ₃			HN ₂ O ₃ [–] = N(H)(O)NO ₂ [–] , hydrido-1κH-trioxido- 1κO,2κ ² O-dinitrate(N–N)(1–)	
HN ₃ O			HON ₃ ^{•–} , hydroxido-1κO- trinitrate(2 N–N)(•1–)	
HO	HO [•] , oxidanyl, hydridoxygen(•); hydroxyl –OH, oxidanyl; hydroxy	HO ⁺ , oxidanylium, hydridoxygen(1+); hydroxylum	HO [–] , oxidanide, hydridoxygenate(1–); hydroxide	HO [–] , oxidanido; hydroxido
HOP	HPO = [P(H)O], phosphanone, hydridoxygenphosphorus >PH(O), oxo-λ ⁵ -phosphanediyl; phosphonoyl = PH(O), oxo-λ ⁵ -phosphanylidene; phosphonoylidene = P–OH, hydroxyphosphanylidene			
HOS	–SH(O), oxo-λ ⁴ -sulfanyl –SOH, hydroxysulfanyl –OSH, sulfanyloxy		HSO [–] , sulfanolate, hydridoxygenate(1–)	HSO [–] , sulfanolato, hydridoxygenate(1–)
HOSe	–SeH(O), oxo-λ ⁴ -selanyl –SeOH, hydroxyselanyl –OSeH, selanyloxy			

Table IX Continued

Formula for uncharged atom or group	Name			
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
HO ₂	HO ₂ [•] , dioxidanyl, hydridodioxygen(•) -OOH, dioxidanyl; hydroperoxy	HO ₂ ⁺ , dioxidanylium, hydridodioxygen(1+)	HO ₂ ⁻ , dioxidanide, hydrogen(peroxide)(1-)	HO ₂ ⁻ , dioxidanido, hydrogen(peroxido)(1-)
HO ₂ P	P(O)(OH), hydroxyphosphanone, hydroxidooxidophosphorus >P(O)(OH), hydroxyoxo-λ ⁵ -phosphanedyl; hydroxyphosphoryl =P(O)(OH), hydroxyoxo-λ ⁵ -phosphanylidene; hydroxyphosphorylidene		HOPO ^{•-} , hydroxidooxidophosphate(•1-) HPO ₂ ²⁻ , hydridodioxidophosphate(2-)	HOPO ^{•-} , hydroxidooxidophosphato(•1-) HPO ₂ ²⁻ , hydridodioxidophosphato(2-)
HO ₂ S	HOOS [•] , hydrido-1κH-sulido-2κS-dioxygen(•) HOSO [•] , hydroxidooxidofur(•) HSOO [•] , (hydridosulfato)dioxygen(•) -S(O)(OH), hydroxyoxo-λ ⁴ -sulfanyl; hydroxysulfinyl, sulfinio -S(O) ₂ H, dioxo-λ ⁶ -sulfanyl		HOSO ⁻ , hydroxysulfanolate, hydroxidooxidofate(1-)	HOSO ⁻ , hydroxysulfanolato, hydroxidooxidofate(1-)

HO ₂ Se	–Se(O)(OH), hydroxyoxo-λ ⁴ -selenyl; hydroxyseleninyl, selenino –Se(O) ₂ H, dioxo-λ ⁶ -selenyl			
HO ₃	HO ₃ [•] , hydriotrioxigen(•) HOOO [•] , trioxidanyl, hydrido- 1κH-trioxygen(2 O–O)(•) –OOOH, trioxidanyl			
HO ₃ P	P(O) ₂ (OH), hydroxy-λ ⁵ -phosphanedione, hydroxidodioxidophosphorus	HOPO ₂ ^{•–} , hydroxidodioxidophosphate(•1–) PHO ₃ ^{2–} , hydriotrioxidophosphate(2–); phosphonate HPO ₃ ^{2–} = PO ₂ (OH) ^{2–} , hydroxidodioxidophosphate(2–); hydrogenphosphite	HOPO ₂ ^{•–} , hydroxidodioxidophosphato(•1–) PHO ₃ ^{2–} , hydriotrioxidophosphato(2–); phosphonate HPO ₃ ^{2–} = PO ₂ (OH) ^{2–} , hydroxidodioxidophosphato(2–); hydrogenphosphito	
HO ₃ S	–S(O) ₂ (OH), hydroxydioxo-λ ⁶ -sulfanyl, hydroxysulfonyl; sulfo		HSO ₃ [–] , hydroxidodioxidosulfate(1–), hydrogensulfite	HSO ₃ [–] , hydroxidodioxidosulfato(1–), hydrogensulfito
HO ₃ Se	HOSeO ₂ [•] , hydroxidodioxidoselenium(•) –Se(O) ₂ (OH), hydroxydioxo-λ ⁶ -selenyl, hydroxyselenonyl; selenono		HSO ₃ [–] , hydroxidodioxidoselenate(1–)	HSO ₃ [–] , hydroxidodioxidoselenato(1–)
HO ₄ P			HOPO ₃ ^{•–} = PO ₃ (OH) ^{•–} , hydroxidotrioxidophosphate(•1–) HPO ₄ ^{2–} = PO ₃ (OH) ^{2–} , hydroxidotrioxidophosphate(2–); hydrogenphosphate	HOPO ₃ ^{•–} = PO ₃ (OH) ^{•–} , hydroxidotrioxidophosphato(•1–) HPO ₄ ^{2–} , hydroxidotrioxidophosphato(2–); hydrogenphosphato
HO ₄ S	HOSeO ₃ [•] , hydro- xidotrioxidosulfur(•) –OS(O) ₂ (OH), hydroxysulfonyloxy; sulfooxy		HSO ₄ [–] , hydroxidotrioxidosulfate(1–); hydrogensulfate	HSO ₄ [–] , hydroxidotrioxidosulfato(1–); hydrogensulfato

Table IX Continued

Formula for uncharged atom or group		Name		
	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
HO ₄ Se			HSeO ₄ ⁻ , hydroxidotrioxoselenate(1-)	HSeO ₄ ⁻ , hydroxidotrioxoselenato(1-)
HO ₅ P			HOPO ₄ ^{•-} = PO ₂ (OH)(OO) ^{•-} , (dioxido)hydroxidodioxido phosphate(•1-)	PO ₂ (OH)(OO) ^{•-} , (dioxido)hydroxidodioxido phosphato(•1-)
HO ₅ S	HOSO ₄ [•] = [SO ₂ (OH)(OO)] [•] , (dioxido)hydroxidodioxidosulfur(•)			
HS	-SH, sulfanyl	HS ⁺ , sulfanylium, hydridosulfur(1+)	HS ⁻ , sulfanide, hydrogen(sulfide)(1-)	HS ⁻ , sulfanido, hydrogen(sulfido)(1-)
HS ₂	HS [•] , sulfanyl, hydridosulfur(•)		HSS ⁻ , disulfanide	HSS ⁻ , disulfanido
HS ₃	-SSH, disulfanyl		HSSS ⁻ , trisulfanide	HSSS ⁻ , trisulfanido
HS ₄	-SSSH, trisulfanyl		HSSSS ⁻ , tetrasulfanide	HSSSS ⁻ , tetrasulfanido
HS ₅	-SSSSH, tetrasulfanyl		HSSSSS ⁻ , pentasulfanide	HSSSSS ⁻ , pentasulfanido
HSe	HSe [•] , selanyl, hydridoselenium(•)	HSe ⁺ , selanylium, hydridoselenium(1+)	HSe ⁻ , selanide, hydrogen(selenide)(1-)	HSe ⁻ , selanido, hydrogen(selenido)(1-)
	-SeH, selanyl		HSeSe ⁻ , diselanide	HSeSe ⁻ , diselanido
HSe ₂	-SeSeH, diselanyl		HTe ⁻ , tellanide, hydrogen(tellanide)(1-)	
HTe	HTe [•] , tellanyl, hydridotellurium(•)	HTe ⁺ , tellanylium, hydridotellurium(1+)		
	-TeH, tellanyl			
HTe ₂	-TeTeH, ditellanyl		HTeTe ⁻ , ditellanide	HTeTe ⁻ , ditellanido
H ₂	H ₂ , dihydrogen	H ₂ ^{•+} , dihydrogen(•1+)		
	D ₂ , dideuterium	¹ H ₂ ^{•+} , diprotium(•1+)		
	T ₂ , ditritium	D ₂ ^{•+} , dideuterium(•1+)		
		T ₂ ^{•+} , ditritium(•1+)		

H_2Br	$H_2Br^\bullet$, λ^3 -bromanyl, dihydrdobromine($\bullet$)	H_2Br^+ , bromanium, dihydrdobromine($1+$)		
H_2Cl	$H_2Cl^\bullet$, λ^3 -chloranyl, dihydrdochlorine($\bullet$)	H_2Cl^+ , chloranium, dihydrdochlorine($1+$)		
H_2F	$H_2F^\bullet$, λ^3 -fluoranyl, dihydrdofluorine($\bullet$)	H_2F^+ , fluoranium, dihydrdofluorine($1+$)		
H_2I	$H_2I^\bullet$, λ^3 -iodanyl, dihydrdoiodine($\bullet$)	H_2I^+ , iodanium, dihydrdoiodine($1+$)		
H_2IO_2	$-I(OH)_2$, dihydroxy- λ^3 -iodanyl			
H_2MnO_4	$H_2MnO_4 = [MnO_2(OH)_2]$, dihydroxidodioxidomanganese			
H_2N_m , see N_mH_2				
H_2NO	$H_2NO^\bullet$, aminooxidanyl, dihydrdooxidonitrogen($\bullet$); aminoxyl $HONH^\bullet$, hydroxyazanyl, hydridohydroxididonitrogen($\bullet$) $-NH(OH)$, hydroxyazanyl, hydroxyamino $-ONH_2$, aminooxy $-NH_2(O)$, oxo- λ^5 -azanyl; azinoyl	$HONH^-$, hydroxyazanide, hydridohydroxididonitrate($1-$) H_2NO^- , azanolate, aminooxidamide, dihydrdooxidonitrate($1-$)	$NHOH^-$, hydroxyazanido, hydridohydroxididonitrate($1-$) H_2NO^- , azanolate, aminooxidamide, dihydrdooxidonitrate($1-$)	
H_2NOS	$-S(O)NH_2$, azanyloxo- λ^4 -sulfanyl; aminosulfanyl			
H_2NO_2S	$-S(O)_2NH_2$, azanyldioxo- λ^6 -sulfanyl; aminosulfonyl; sulfamoyl			
H_2NO_3		$[NO(OH)_2]^+$, dihydroxidooxididonitrogen($1+$)		
H_2NS	$-SNH_2$, azanyl/sulfanyl; aminosulfanyl $-NH_2(S)$, sulfanylidene- λ^5 -azanyl; azinothioyl			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
H_2N_m , see N_mH_2				
H_2O	H_2O , dihydrogen oxide; water $\text{H}_2\text{O} = [\text{OH}_2]$, oxidane (parent hydride name), dihydrodooxygen $^1\text{H}_2\text{O}$, diprotium oxide; $(^1\text{H}_2)\text{water}$ $\text{D}_2\text{O} = ^2\text{H}_2\text{O}$, dideuterium oxide; $(^2\text{H}_2)\text{water}$ $\text{T}_2\text{O} = ^3\text{H}_2\text{O}$, ditritium oxide; $(^3\text{H}_2)\text{water}$			H_2O , aqua
H_2OP	$-\text{PH}_2\text{O}$, oxo- λ^5 -phosphanyl; phosphinoyl		PH_2O^- dihydrodooxidophosphate(1-); phosphinite	PH_2O^- , dihydrodooxidophosphato(1-); phosphinito
H_2OSb	$-\text{SbH}_2\text{O}$, oxo- λ^5 -stibanyl, stibinoyl			
H_2O_2	H_2O_2 , dihydrogen peroxide; hydrogen peroxide HOOH , dioxidane (parent hydride name), bis(hydrodooxygen)($O-O$)	$\text{HOOH}^{\bullet+}$, dioxidaniumyl, bis(hydrodooxygen)($O-O$)($\bullet 1^+$)		HOOH , dioxidane
$\text{H}_2\text{O}_2\text{P}$	$-\text{P}(\text{OH})_2$, dihydroxyphosphanyl $-\text{PH}(\text{O})(\text{OH})$, hydroxyoxo- λ^5 -phosphanyl		PH_2O_2^- , dihydrodooxidophosphate(1-); phosphinate	PH_2O_2^- , dihydrodooxidophosphato(1-); phosphinato

H_2O_3B				$H_2BO_3^- = [BO(OH)_2]^-$, dihydroxidooxidoborate(1-); dihydrogenborate	$H_2BO_3^- = [BO(OH)_2]^-$, dihydroxidooxidoborate(1-); dihydrogenborato
H_2O_3P	$-P(O)(OH)_2$, dihydroxyoxo- λ^5 -phosphanyl; dihydroxyphosphoryl, phosphono			$[PHO_2(OH)]^-$, hydrihydroxidodioxido- phosphate(1-); hydrogenphosphonate $[PO(OH)_2]^-$, dihydroxidooxidophosphate(1-); dihydrogenphosphite	$[PHO_2(OH)]^-$, hydrihydroxidodioxido- phosphato(1-); hydrogenphosphonato $[PO(OH)_2]^-$, dihydroxidooxidophosphato(1-); dihydrogenphosphito
H_2O_4P	$(HO)_2PO_2^\bullet$, (dihydroxido)dioxidophosphorus($\bullet$)			$H_2PO_4^-$, dihydroxidodioxidophosphate(1-); dihydrogenphosphate	$H_2PO_4^-$, dihydroxidodioxidophosphato(1-); dihydrogenphosphato
$H_2O_5P_2$				$P_2H_2O_5^{2-} = [PH(O)_2OPH(O)_2]^{2-}$, μ -oxido- bis(hydriodioxidophosphate)(2-); diphosphonate	$P_2H_2O_5^{2-} = [PH(O)_2OPH(O)_2]^{2-}$, μ -oxido- bis(hydriodioxidophosphato)(2-); diphosphonato
H_2PS	$-PH_2(S)$, sulfanylidene- λ^5 -phosphanyl; phosphinothioyl				
H_2Po	H_2Po , dihydrogen polonide $H_2Po = [PoH_2]$, polane (parent hydride name), dihydriodopolonium				
H_2S	H_2S , dihydrogen sulfide; hydrogen sulfide $H_2S = [SH_2]$, sulfane (parent hydride name), dihydridosulfur	$H_2S^{\bullet+}$, sulfaniumyl, dihydridosulfur($\bullet 1+$) $-SH_2^+$, sulfaniumyl		$H_2S^{\bullet-}$, sulfaniumyl, dihydridosulfate($\bullet 1-$) ^e	H_2S , sulfane
H_2S_2	H_2S_2 , dihydrogen disulfide HSSH, disulfane (parent hydride name), bis(hydridosulfur)($S-S$)	$HSSH^{\bullet+}$, disulfaniumyl, bis(hydridosulfur)($S-S$)($\bullet 1+$)		$HSSH^{\bullet-}$, disulfaniumyl, bis(hydridosulfate)($S-S$)($\bullet 1-$) ^e	HSSH, disulfane
H_2S_3	H_2S_3 , dihydrogen trisulfide HSSSH, trisulfane (parent hydride name)				HSSSH, trisulfane

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
H ₂ S ₄	H ₂ S ₄ , dihydrogen tetrasulfide HSSSSH, tetrasulfane (parent hydride name)			HSSSSH, tetrasulfane
H ₂ S ₅	H ₂ S ₅ , dihydrogen pentasulfide HSSSSSH, pentasulfane (parent hydride name)			HSSSSSH, pentasulfane
H ₂ Se	H ₂ Se, dihydrogen selenide; hydrogen selenide H ₂ Se = [SeH ₂], selane (parent hydride name), dihydridoselenium	H ₂ Se ^{•+} , seleniumyl, dihydridoselenium(•1+) –SeH ₂ ⁺ , seleniumyl	H ₂ Se ^{•–} , selenuidyl, dihydridoselenate(•1–) ^e	H ₂ Se, selane
H ₂ Se ₂	H ₂ Se ₂ , dihydrogen diselenide HSeSeH, diselane (parent hydride name), bis(hydridoselenium)(Se–Se)	HSeSeH ^{•+} , diselaniumyl, bis(hydridoselenium)(Se–Se)(•1+)	HSeSeH ^{•–} , diselanuidyl, bis(hydridoselenate)(Se–Se)(•1–) ^e	HSeSeH, diselane
H ₂ Te	H ₂ Te, dihydrogen tellanide; hydrogen tellanide H ₂ Te = [TeH ₂], tellane (parent hydride name), dihydridotellurium	H ₂ Te ^{•+} , tellaniumyl, dihydridotellurium(•1+) –TeH ₂ ⁺ , tellaniumyl	H ₂ Te ^{•–} , tellanuidyl, dihydridotellurate(•1–) ^e	H ₂ Te, tellane
H ₃		H ₃ ⁺ , trihydrogen(1+)		
H ₃ N _m , see N _m H ₃				

H_3NO	HONH_2 , azanol, dihydrodihydroxidodinitrogen; hydroxylamine (parent name for organic derivatives)	$\text{HONH}_2^{\bullet+}$, hydroxyazanumyl, dihydrodihydroxidodinitrogen($\bullet 1+$)		HONH_2 , azanol, dihydrodihydroxidodinitrogen; hydroxylamine
H_3NP	$-\text{PH}_2(=\text{NH})$, imino- λ^5 -phosphanyl; phosphinimidoyl			
H_3O		H_3O^+ , oxidanium, trihydrodooxygen($1+$), aquahydrogen($1+$); oxonium (<i>not</i> hydronium)		
H_3OS	$\text{H}_3\text{OS}^+ = [\text{SH}_3(\text{O})]^+$, oxo- λ^5 -sulfanylium, trihydrodooxidosulfur($1+$)			
H_3OSi	$-\text{OSiH}_3$, silyloxy			
$\text{H}_3\text{O}_4\text{S}$		$[\text{SO}(\text{OH})_3]^+ = \text{H}_3\text{SO}_4^+$, trihydrodooxidosulfur($1+$), trihydrogen(tetraoxidosulfate)($1+$)		
$\text{H}_3\text{O}_5\text{P}$	$[\text{PO}(\text{OH})_2(\text{OOH})]$, (dioxidanido)dihydroxido $\ominus$ oxidophosphorus; peroxyphosphoric acid, phosphoroperoxoic acid			
H_3S	$\text{H}_3\text{S}^\bullet$, λ^4 -sulfanyl, trihydrodisulfur($\bullet$)	H_3S^+ , sulfanium, trihydrodisulfur($1+$)	H_3S^- , sulfanuide, trihydrodisulfate($1-$)	
H_3Se	$\text{H}_3\text{Se}^\bullet$, λ^4 -selanyl, trihydrodiselenium($\bullet$)	H_3S^+ , selanium, trihydrodiselenium($1+$)	H_3Se^- , selanuide, trihydrodiselenate($1-$)	
H_3Te	$\text{H}_3\text{Te}^\bullet$, λ^4 -tellanyl, trihydrodote tellurium($\bullet$)	H_3Te^+ , tellanium, trihydrodote tellurium($1+$)	H_3Te^- , tellanuide, trihydrodote tellurate($1-$)	
H_4N_m , see N_mH_4				
H_4NO		NH_2OH_2^+ , aminooxidanium, aquadihydroxidodinitrogen($1+$) NH_3OH^+ , hydroxyazanum, trihydrodihydroxidodinitrogen($1+$); hydroxyammonium		

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
H ₄ O		H ₄ O ²⁺ , oxidanadium, tetrahydridoxygen(2+)		
H ₅ IO ₆	IO(OH) ₅ , pentahydroxy-λ ⁷ -iodanone, pentahydroxidoiodine; orthoperiodic acid			
H ₅ N ₂ , see N ₂ H ₅				
H ₅ O ₂		[H(H ₂ O) ₂] ⁺ , μ-hydrido- bis(dihydrodooxygen)(1+), diaquahydrogen(1+)		
H ₆ N ₂ , see N ₂ H ₆				
H _n N _m , see N _m H _n				
He	helium	helium (general) He ^{•+} , helium(•1+)	helide	helido
HeH		HeH ⁺ , hydridohelium(1+)		
He ₂		He ₂ ⁺ , dihelium(1+) He ₂ ²⁺ , dihelium(2+)		
Hf	hafnium	hafnium	hafnide	hafnido
Hg	mercury	mercury (general) Hg ²⁺ , mercury(2+)	mercuride	mercurido
Hg ₂		Hg ₂ ²⁺ , dimercury(2+)		
Ho	holmium	holmium	holmide	holmido

Hs	hassium	hassium	hassium	hassido
I	iodine (general) I [•] , iodine(•), monoiodine –I, iodo	iodine (general) I ⁺ , iodine(1+)	iodide (general) I [–] , iodide(1 [–]); iodide	I [–] , iodido(1 [–]); iodido
ICl ₂	ICl ₂ [•] , dichloroiodine(•) –ICl ₂ , dichloro-λ ³ -iodanyl	ICl ₂ ⁺ , dichloroiodanium, dichloroiodine(1+)		
IF	IF, iodine fluoride [IF], fluoroiodine			
IF ₄		IF ₄ ⁺ , tetrafluoro-λ ³ -iodanium, tetrafluoroiodine(1+)	IF ₄ [–] , tetrafluoro-λ ³ -iodanuide, tetrafluoroiodate(1 [–])	IF ₄ [–] , tetrafluoro-λ ³ -iodanuïdo, tetrafluoroiodato(1 [–])
IF ₆			IF ₆ [–] , hexafluoro-λ ⁵ -iodanuide, hexafluoroiodate(1 [–])	IF ₆ [–] , hexafluoro-λ ⁵ -iodanuïdo, hexafluoroiodato(1 [–])
IH, see HI				
I ₂	I ₂ , diiodine	I ₂ ^{•+} , diiodine(•1+)	I ₂ ^{•–} , diiodide(•1 [–])	I ₂ , diiodine
I ₃	I ₃ , triiodine		I ₃ [–] , triiodide(1 [–]); triiodide	I ₃ [–] , triiodido(1 [–]); triiodido
In	indium	indium	indide	indido
InH ₂	–InH ₂ , indiganyl			
InH ₃	InH ₃ , indium trihydride [InH ₃], indigane (parent hydride name), trihydridoindium			
Ir	iridium	iridium	iridide	iridido
K	potassium	potassium	potasside	potassido
KO ₂	KO ₂ , potassium dioxide(1 [–]); potassium superoxide			
KO ₃	KO ₃ , potassium trioxide(1 [–]); potassium ozonide			
K ₂ O	K ₂ O, dipotassium oxide			
K ₂ O ₂	K ₂ O ₂ , dipotassium dioxide(2 [–]); potassium peroxide			
Kr	krypton	krypton	kryptonide	kryptonido
La	lanthanum	lanthanum	lanthanide ^d	lanthanido
Li	lithium	lithium (general) Li ⁺ , lithium(1+)	lithide (general) Li [–] , lithide(1 [–]); lithide	lithido Li [–] , lithido(1 [–]); lithido

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
LiAl	[LiAl], aluminidolithium			
LiBe	[LiBe], beryllidolithium			
LiCl	LiCl, lithium chloride [LiCl], chloridolithium	LiCl ⁺ , chloridolithium(1+)	LiCl ⁻ , chloridolithate(1-)	LiCl ⁻ , chloridolithato(1-)
LiH	LiH, lithium hydride [LiH], hydridolithium	LiH ⁺ , hydridolithium(1+)	LiH ⁻ , hydridolithate(1-)	LiH ⁻ , hydridolithato(1-)
LiMg	LiMg, lithium monomagneside	LiMg ⁺ , magnesidolithium(1+)		
Li ₂	Li ₂ , dilithium	Li ₂ ⁺ , dilithium(•1+)	Li ₂ ⁻ , dilithide(•1-)	Li ₂ ⁻ , dilithido(•1-)
Lr	lawrencium	lawrencium	lawrencide	lawrencido
Lu	lutetium	lutetium	lutetide	lutetido
Md	mendelevium	mendelevium	mendelevide	mendelevido
Mg	magnesium	magnesium (general) Mg ⁺ , magnesium(1+) Mg ²⁺ , magnesium(2+)	magneside (general) Mg ⁻ , magneside(1-)	magnesido Mg ⁻ , magnesido(1-)
Mn	manganese	manganese (general) Mn ²⁺ , manganese(2+) Mn ³⁺ , manganese(3+)	manganide	manganido
MnO	MnO, manganese mon(o)oxide, manganese(II) oxide			
MnO ₂	MnO ₂ , manganese dioxide, manganese(IV) oxide			
MnO ₃		MnO ₃ ⁺ , trioxidomanganese(1+)		

MnO ₄				MnO ₄ ⁻ , tetraoxidomanganate(1-), permanganate MnO ₄ ²⁻ , tetraoxidomanganate(2-), manganate(VI) MnO ₄ ³⁻ , tetraoxidomanganate(3-), manganate(V)	MnO ₄ ⁻ , tetraoxidomanganato(1-), permanganato MnO ₄ ²⁻ , tetraoxidomanganato(2-), manganato(VI) MnO ₄ ³⁻ , tetraoxidomanganato(3-), manganato(V)
Mn ₂ O ₃	Mn ₂ O ₃ , dimanganese trioxide, manganese(III) oxide				
Mn ₂ O ₇	Mn ₂ O ₇ , dimanganese heptaoxide, manganese(VII) oxide [O ₃ MnOMnO ₃], μ-oxido-bis(trioxidomanganese)				
Mn ₃ O ₄	Mn ₃ O ₄ , trimanganese tetraoxide Mn ^{II} Mn ^{III} ₂ O ₄ , manganese(II,III) tetraoxide				
Mo	molybdenum	molybdenum	molybdenide	molybdenido	molybdenido
Mt	meitnerium	meitnerium	meitneride	meitnerido	meitnerido
Mu	Mu [•] = μ ⁺ e ⁻ , muonium	Mu ⁺ = μ ⁺ , muon	Mu ⁻ = μ ⁺ (e ⁻) ² , muonide		
N	nitrogen N [•] , nitrogen(•), mononitrogen -N<, azanetriyl; nitrilo -N=, azanylylidene ≡N, azanylidyne	nitrogen (general) N ⁺ , nitrogen(1+)	nitride (general) N ³⁻ , nitride(3-), azanetriide; nitride =N-, azanidylydene; amidylidene -N ²⁻ , azanediidyl	N ³⁻ , nitrido(3-), azanetriido	
NCO, see CNO					
NCS, see CNS					
NCl ₂			NCl ₂ ⁻ , dichloroazanide, dichloridonitrate(1-)	NCl ₂ ⁻ , dichloroazanido, dichloridonitrato(1-)	
NF			NF ²⁻ , fluoroazanediide, fluoridonitrate(2-)	NF ²⁻ , fluoroazanediido, fluoridonitrato(2-)	

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
NF ₃	NF ₃ , nitrogen trifluoride [NF ₃], trifluoroazane, trifluoridonitrogen			NF ₃ , trifluoroazane, trifluoridonitrogen
NF ₄		NF ₄ ⁺ , tetrafluoroammonium, tetrafluoroazanium, tetrafluoridonitrogen(1+)		
NH	NH ^{2•} , azanylidene, hydridonitrogen(2•); nitrene >NH, azanediyl =NH, azanylidene; imino	NH ⁺ , azanyliumdiyl, hydridonitrogen(1+) NH ²⁺ , azanebis(ylium), hydridonitrogen(2+)	NH ⁻ , azanidyl, hydridonitrate(1-) NH ²⁻ , azanediide, hydridonitrate(2-); imide -NH ⁻ , azanidyl; amidyl	NH ²⁻ , azanediido, hydridonitrate(2-); imido
NH ₂	NH ₂ [•] , azanyl, dihydridonitrogen(•); aminyl -NH ₂ , azanyl; amino	NH ₂ ⁺ , azanylium, dihydridonitrogen(1+)	NH ₂ ⁻ , azanide, dihydridonitrate(1-); amide	NH ₂ ⁻ , azanido, dihydridonitrate(1-), amido
NH ₃	NH ₃ , azane (parent hydride name), amine (parent name for certain organic derivatives), trihydridonitrogen; ammonia	NH ₃ ^{•+} , azaniumyl, trihydridonitrogen(•1+) -NH ₃ ⁺ , azaniumyl; ammonio	NH ₃ ^{•-} , azanuidyl, trihydridonitrate(•1-) ^e	NH ₃ , ammine
NH ₄	NH ₄ [•] , λ ⁵ -azanyl, tetrahydridonitrogen(•)	NH ₄ ⁺ , azanium; ammonium		

NO	NO, nitrogen mon(o)oxide (<i>not</i> nitric oxide) NO [•] , oxoazanyl, oxidonitrogen(•); nitrosyl -N=O, oxoazanyl; nitroso >N(O)-, oxo- λ^3 -azanetriyl; azoryl =N(O)-, oxo- λ^5 -azanylidene; azorylidene $\equiv$ N(O), oxo- λ^5 -azanylidene; azorylidene -O ⁺ =N ⁻ , azanidyldeneoxoazaniumyl	NO ⁺ , oxidonitrogen(1+) (<i>not</i> nitrosyl) NO ^{•2+} , oxidonitrogen(•2+)	NO ⁻ , oxidonitrate(1-) NO ^(2•-) , oxidonitrate(•2-1-)	NO, oxidonitrogen (general); nitrosyl = oxidonitrogen-κN (general) NO ⁺ , oxidonitrogen(1+) NO ⁻ , oxidonitrate(1-)
NO ₂	NO ₂ , nitrogen dioxide NO ₂ [•] = ONO [•] , nitrosooxidanyl, dioxidonitrogen(•); nitryl -NO ₂ , nitro -ONO, nitrosoxy	NO ₂ ⁺ , dioxidonitrogen(1+) (<i>not</i> nitryl)	NO ₂ ⁻ , dioxidonitrate(1-); nitrite NO ₂ ^{•2-} , dioxidonitrate(•2-)	NO ₂ ⁻ , dioxidonitrate(1-); nitrite NO ₂ ^{•2-} , dioxidonitrate(•2-)
NO ₃	NO ₃ , nitrogen trioxide NO ₃ [•] = O ₂ NO [•] , nitrooxidanyl, trioxidonitrogen(•) ONOO [•] , nitrosodioxidanyl, oxidoperoxidonitrogen(•) -ONO ₂ , nitrooxy		NO ₃ ⁻ , trioxidonitrate(1-); nitrate NO ₃ ^{•2-} , trioxidonitrate(•2-) [NO(OO)] ⁻ , oxidoperoxidonitrate(1-); peroxynitrite	NO ₃ ⁻ , trioxidonitrate(1-); nitrate NO ₃ ^{•2-} , trioxidonitrate(•2-) [NO(OO)] ⁻ , oxidoperoxidonitrate(1-); peroxynitrite
NO ₄			NO ₂ (O ₂) ⁻ , dioxidoperoxidonitrate(1-); peroxynitrate	NO ₂ (O ₂) ⁻ , dioxidoperoxidonitrate(1-); peroxynitrate
NS	NS, nitrogen monosulfide NS [•] , sulfidonitrogen(•) -N=S, sulfanylideneazanyl; thionitroso	NS ⁺ , sulfidonitrogen(1+) (<i>not</i> thionitrosyl)	NS ⁻ , sulfidonitrate(1-)	NS, sulfidonitrogen, sulfidonitrate, thionitrosyl (general) NS ⁺ , sulfidonitrogen(1+) NS ⁻ , sulfidonitrate(1-)

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
N ₂	N ₂ , dinitrogen =N ⁺ =N ⁻ , (azanidyliene)azanumylidene; diazo =NN=, diazene-1,2-diylidene; hydrazinediylidene -N=N-, diazene-1,2-diyl; azo	N ₂ ^{•+} , dinitrogen(•1+) N ₂ ²⁺ , dinitrogen(2+) -N ⁺ ≡N, diazyn-1-ium-1-yl	N ₂ ²⁻ , dinitride(2-) N ₂ ⁴⁻ , dinitride(4-), diazanetetraide; hydrazinetetraide	N ₂ , dinitrogen N ₂ ²⁻ , dinitride(2-) N ₂ ⁴⁻ , dinitride(4-), diazanetetraide; hydrazinetetraide
N ₂ H		N≡NH ⁺ , diazynium	N=NH ⁻ , diazenide NNH ³⁻ , diazanetriide, hydrazinetriide	N=NH ⁻ , diazenido NNH ³⁻ , diazanetriido, hydrazinetriido
N ₂ H ₂	HN=NH, diazene N=NH ₂ ⁺ , diazen-2-ium-1-ide H ₂ NN ^{2•} , diazanylidene, hydrazinylidene =NNH ₂ , diazanylidene; hydrazinylidene •HNNH [•] , diazane-1,2-diyl; hydrazine-1,2-diyl -HNNH-, diazane-1,2-diyl; hydrazine-1,2-diyl	HNNH ²⁺ , diazynylium	HNNH ²⁻ , diazane-1,2-diide, hydrazine-1,2-diide H ₂ NN ²⁻ , diazane-1,1-diide, hydrazine-1,1-diide	HN=NH, diazene N=NH ₂ ⁺ , diazen-2-ium-1-ido HNNH ²⁻ , diazane-1,2-diido, hydrazine-1,2-diido H ₂ NN ²⁻ , diazane-1,1-diido, hydrazine-1,1-diido
N ₂ H ₃	H ₂ NNH [•] , diazanyl, trihydridodinitrogen(N—N)(•); hydrazinyl -NHNH ₂ , diazanyl; hydrazinyl 2-NNH ₃ ⁺ , diazan-2-ium-1,1-diide	HN=NH ₂ ⁺ , diazenium	H ₂ NNH ⁻ , diazanide, hydrazinide	2-NNH ₃ ⁺ , diazan-2-ium-1,1-diido H ₂ NNH ⁻ , diazanido, hydrazinido

N_2H_4	H_2NNH_2 , diazane (parent hydride name), hydrazine (parent name for organic derivatives) $\overline{N}NHNH_3^+$, diazan-2-ium-1-ido	$H_2NNH_2^{\bullet+}$, diazaniumyl, bis(dihydridonitrogen)($N-N$)($\bullet 1+$); hydraziniumyl $H_2N=NH_2^{2+}$, diazenedidium $H_2NNH_3^+$, diazanium, hydrazinium	H_2NNH_2 , diazane, hydrazine $\overline{N}NHNH_3^+$, diazan-2-ium-1-ido
N_2H_5			
N_2H_6		$H_3NNH_3^{2+}$, diazanedidium, hydrazinedidium	
N_2O	N_2O , dinitrogen oxide (<i>not</i> nitrous oxide) NNO, oxidodinitrogen($N-N$) $-N(O)=N-$, azoxy	$N_2O^{\bullet-}$, oxidodinitrate($\bullet 1-$)	N_2O , dinitrogen oxide (general) NNO, oxidodinitrogen($N-N$) $N_2O^{\bullet-}$, oxidodinitrate($\bullet 1-$)
N_2O_2	N_2O_2 , dinitrogen dioxide ONNO, bis(oxidonitrogen)($N-N$)		$N_2O_2^{2-}$, diazenediolate, bis(oxidonitrate)($N-N$)(2-)
N_2O_3	N_2O_3 , dinitrogen trioxide O_2NNO , trioxido-1 $\kappa^2O_2,2\kappa O$ -dinitrogen($N-N$) $NO^+NO_2^-$, oxidonitrogen(1+) dioxidinitrate(1-) ONONO, dinitrosooxidane, μ -oxido-bis(oxidonitrogen)	$N_2O_3^{2-} = [O_2NNO]^{2-}$, trioxido-1 $\kappa^2O_2,2\kappa O$ -dinitrate($N-N$)(2-)	
N_2O_4	N_2O_4 , dinitrogen tetraoxide O_2NNO_2 , bis(dioxidonitrogen)($N-N$) ONONO, 1,2-dinitrosodioxidane, bis(nitrosyloxygen)($O-O$), 2,5-diazy-1,3,4,6-tetraoxy-[6]catena $NO^+NO_3^-$, oxidonitrogen(1+) trioxidinitrate(1-)		

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
N ₂ O ₅	N ₂ O ₅ , dinitrogen pentaoxide O ₂ NONO ₂ , dinitrooxidane, μ-oxido- bis(dioxidonitrogen)(N—N) NO ₂ ⁺ NO ₃ [−] , dioxidonitrogen(1+) trioxidonitrate(1−)			
N ₃	N ₃ [•] , trinitrogen(•) −N≡N ⁺ =N [−] , azido		N ₃ [−] , trinitride(1−); azide	N ₃ [−] , trinitrido(1−); azido
N ₃ H	N ₃ H, hydrogen trinitride(1−); hydrogen azide [NNNH], hydrido-1κ <i>H</i> -trinitrogen(2 <i>N</i> — <i>N</i>)			
N ₃ H ₂	−NHN=NH, triaz-2-en-1-yl			
N ₃ H ₄	−NHNHNH ₂ , triazan-1-yl			
N ₅		N ₅ ⁺ , pentanitrogen(1+)		
N ₆			N ₆ ^{•−} , hexanitride(•1−)	N ₆ ^{•−} , hexanitrido(•1−)
Na	sodium	sodium (general) Na ⁺ , sodium(1+)	sodide (general) Na [−] , sodide(1−); sodide	sodido Na [−] , sodido(1−); sodido
NaCl	NaCl, sodium chloride [NaCl], chloridosodium	NaCl ¹⁺ , chloridosodium(1+)	NaCl ^{1−} , chloridosodate(1−)	
Na ₂	Na ₂ , disodium	Na ₂ ⁺ , disodium(1+)	Na ₂ [−] , disodide(1−)	Na ₂ [−] , disodido(1−)
Nb	niobium	niobium	niobide	niobido
Nd	neodymium	neodymium	neodymide	neodymido

Ne	neon	neon (general) Ne ⁺ , neon(1+)	neonide	neonido
NeH		NeH ⁺ , hydridoneon(1+)		
NeHe		NeHe ⁺ , helidoneon(1+)		
Ni	nickel	nickel (general) Ni ²⁺ , nickel(2+) Ni ³⁺ , nickel(3+)	nickelide	nickelido
No	nobelium	nobelium	nobelide	nobelido
Np	neptunium	neptunium	neptunide	neptunido
NpO ₂	NpO ₂ , neptunium dioxide	NpO ₂ ⁺ , dioxidoneptunium(1+) [<i>not</i> neptunyl(1+)] NpO ₂ ²⁺ , dioxidoneptunium(2+) [<i>not</i> neptunyl(2+)]		
O	oxygen (general) O, monooxygen O ^{2•} , oxidanylidene, monooxygen(2•) >O, oxy, epoxy (in rings) =O, oxo	oxygen (general) O ^{•+} , oxygen(•1+)	oxide (general) O ^{•-} , oxidanidyl, oxide(•1-) O ²⁻ , oxide(2-); oxide -O ⁻ , oxido	O ²⁻ , oxido
OBr	OBr, oxygen (mono)bromide ^f OBr [•] , bromidooxygen(•) ^f ; bromosyl -BrO, oxo-λ ³ -bromanyl; bromosyl -OBr, bromooxy	OBr ⁺ , bromidooxygen(1+) ^f (<i>not</i> bromosyl)	OBr ⁻ , bromidooxygenate(1-) ^f ; oxidobromate(1-) ^f , hypobromite	OBr ⁻ , bromidooxygenato(1-) ^f ; oxidobromato(1-) ^f , hypobromito
OCN, see CNO				
OCl	OCl, oxygen (mono)chloride ^f OCl [•] , chloridooxygen(•) ^f ; chlorosyl -ClO, oxo-λ ³ -chloranyl; chlorosyl -OCl, chlorooxy		OCl ⁻ , chloridooxygenate(1-) ^f ; oxidochlorate(1-) ^f , hypochlorite	OCl ⁻ , chloridooxygenato(1-) ^f ; oxidochlorato(1-) ^f , hypochlorito
OD ₂ , see H ₂ O				

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
OF	OF, oxygen (mono)fluoride OF [•] , fluoridooxygen(•) -FO, oxo-λ ³ -fluoranyl; fluorosyl	OF ⁺ , fluoridooxygen(1+)	OF ⁻ , fluoridooxygenate(1-)	
OF ₂	OF ₂ , oxygen difluoride [OF ₂], difluoridooxygen			
OH _n , see H _n O (n = 1-4)				
O ¹ H ₂ , see H ₂ O				
OI	OI, oxygen (mono)iodide ^f OI [•] , iodidooxygen(•) ^f ; iodosyl -IO, oxo-λ ³ -iodanyl; iodosyl -OI, iodoxy	OI ⁺ , iodidooxygen(1+) ^f (not iodosyl)	OI ⁻ , iodidooxygenate(1-) ^f ; oxidoiodate(1-) ^f ; hypiodite OI ^{•2-} , iodidooxygenate(•2-) ^f	OI ⁻ , iodidooxygenato(1-) ^f ; oxidoiodato(1-) ^f ; hypiodito
ONC, see CNO				
OT ₂ , see H ₂ O				
O ₂	O ₂ , dioxygen O ₂ ^{2•} , dioxidanediyl, dioxygen(2•) -OO-, dioxidanediyl; peroxy	O ₂ ^{•+} , dioxidanilylumyl, dioxygen(•1+) O ₂ ²⁺ , dioxidanebis(ylium), dioxygen(2+)	O ₂ ^{•-} , dioxidanidyl, dioxide(•1-); superoxide (not hyperoxide) O ₂ ²⁻ , dioxidanediide, dioxide(2-); peroxide	dioxido (general) O ₂ , dioxygen O ₂ ^{•-} , dioxide(•1-); superoxido O ₂ ²⁻ , dioxidanediido, dioxido(2-); peroxido
O ₂ Br	O ₂ Br, dioxygen bromide ^f BrO ₂ [•] , dioxidobromine(•) -BrO ₂ , dioxo-λ ⁵ -bromanyl; bromyl -OBrO, oxo-λ ³ -bromanyloxy	BrO ₂ ⁺ , dioxidobromine(1+) (not bromyl)	BrO ₂ ⁻ , dioxidobromate(1-); bromite	BrO ₂ ⁻ , dioxidobromato(1-); bromito

O_2Cl	O_2Cl , dioxygen chloride ^f $ClO_2^\bullet$, dioxidochlorine($\bullet$) $ClOO^\bullet$, chloridodioxygen($O-O$)($\bullet$) - ClO_2 , dioxo- λ^5 -chloranyl; chloryl - $OCIO$, oxo- λ^3 -chloranyloxy	ClO_2^+ , dioxidochlorine(1+) (<i>not</i> chloryl)	ClO_2^- , dioxidochlorate(1-); chlorite	ClO_2^- , dioxidochlorato(1-); chlorito
O_2Cl_2		$O_2Cl_2^+$, (dioxygen dichloride)(1+) ^f		
O_2F_2	O_2F_2 , dioxygen difluoride FOOF, difluorodioxidane, bis(fluoridooxygen)($O-O$)			
O_2I	O_2I , dioxygen iodide ^f $IO_2^\bullet$, dioxidoiodine($\bullet$) - IO_2 , dioxo- λ^5 -iodanyl; iodyl - OIO , oxo- λ^3 -iodanyloxy	IO_2^+ , dioxidoiodine(1+) (<i>not</i> iodyl)	IO_2^- , dioxidoiodate(1-); iodite	IO_2^- , dioxidoiodato(1-); iodito
O_3	O_3 , trioxxygen; ozone - $OOO-$, trioxidanediyl		$O_3^\bullet$, trioxidanidyl, trioxide($\bullet$ 1-); ozonide	O_3 , trioxxygen; ozone $O_3^\bullet$, trioxido($\bullet$ 1-); ozonido
O_3Br	O_3Br , trioxxygen bromide ^f $BrO_3^\bullet$, trioxidobromine($\bullet$) - BrO_3 , trioxo- λ^7 -bromanyl; perbromyl - $OBRO_2$, dioxo- λ^5 -bromanyloxy	BrO_3^+ , trioxidobromine(1+) (<i>not</i> perbromyl)	BrO_3^- , trioxidobromate(1-); bromate	BrO_3^- , trioxidobromato(1-); bromato
O_3Cl	O_3Cl , trioxxygen chloride ^f $ClO_3^\bullet$, trioxidochlorine($\bullet$) - ClO_3 , trioxo- λ^7 -chloranyl; perchloryl - $OCIO_2$, dioxo- λ^5 -chloranyloxy	ClO_3^+ , trioxidochlorine(1+) (<i>not</i> perchloryl)	ClO_3^- , trioxidochlorate(1-); chlorate	ClO_3^- , trioxidochlorato(1-); chlorato
O_3I	O_3I , trioxxygen iodide ^f $IO_3^\bullet$, trioxidoiodine($\bullet$) - IO_3 , trioxo- λ^7 -iodanyl; periodyl - OIO_2 , dioxo- λ^5 -iodanyloxy	IO_3^+ , trioxidoiodine(1+) (<i>not</i> periodyl)	IO_3^- , trioxidoiodate(1-); iodate	IO_3^- , trioxidoiodato(1-); iodato
O_4Br	O_4Br , tetraoxxygen bromide ^f $BrO_4^\bullet$, tetraoxidobromine($\bullet$) - $OBRO_3$, trioxo- λ^7 -bromanyloxy		BrO_4^- , tetraoxidobromate(1-); perbromate	BrO_4^- , tetraoxidobromato(1-); perbromato

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
O ₄ Cl	O ₄ Cl, tetraoxygen chloride ^f ClO ₄ •, tetraoxidochlorine(•) -OCIO ₃ , trioxo-λ ⁷ -chloranyl oxy		ClO ₄ ⁻ , tetraoxidochlorate(1-); perchlorate	ClO ₄ ⁻ , tetraoxidochlorato(1-); perchlorato
O ₄ I	O ₄ I, tetraoxygen iodide ^f IO ₄ •, tetraoxidoiodine(•) -OIO ₃ , trioxo-λ ⁷ -iodanyl oxy		IO ₄ ⁻ , tetraoxidoiodate(1-); periodate	IO ₄ ⁻ , tetraoxidoiodato(1-); periodato
O ₅ I			IO ₅ ³⁻ , pentaaxidoiodate(3-)	IO ₅ ³⁻ , pentaaxidoiodato(3-)
O ₆ I			IO ₆ ⁵⁻ , hexaaxidoiodate(5-); orthoperiodate	IO ₆ ⁵⁻ , hexaaxidoiodato(5-); orthoperiodato
O ₉ I ₂			I ₂ O ₉ ⁴⁻ , nonaaxidodiiodate(4-) [O ₃ I(μ-O) ₃ IO ₃] ⁴⁻ , tri-μ-oxido-bis(trioxidoiodate)(4-)	I ₂ O ₉ ⁴⁻ , nonaaxidodiiodato(4-) [O ₃ I(μ-O) ₃ IO ₃] ⁴⁻ , tri-μ-oxido-bis(trioxidoiodato)(4-)
Os	osmium	osmium	osmide	osmido
P	phosphorus (general) P•, phosphorus(•), monophosphorus >P-, phosphanetriyl	phosphorus (general) P ⁺ , phosphorus(1+)	phosphide (general) P ⁻ , phosphide(1-) P ³⁻ , phosphide(3-), phosphanetriide; phosphide	P ³⁻ , phosphido, phosphanetriido
PF			PF ²⁻ , fluorophosphanediide, fluoridophosphate(2-)	PF ²⁻ , fluorophosphanediido, fluoridophosphato(2-)
PF ₂			PF ₂ ⁻ , difluorophosphanide, difluoridophosphate(1-)	PF ₂ ⁻ , difluorophosphanido, difluoridophosphato(1-)
PF ₃	PF ₃ , phosphorus trifluoride [PF ₃], trifluorophosphane, trifluoridophosphorus			

PF ₄		PF ₄ ⁺ , tetrafluorophosphanium, tetrafluoridophosphorus(1+)	PF ₄ ⁻ , tetrafluorophosphanuide, tetrafluoridophosphate(1-)	PF ₄ ⁻ , tetrafluorophosphanuido, tetrafluoridophosphato(1-)
PF ₅	PF ₅ , phosphorus pentafluoride [PF ₅], pentafluoro-λ ⁵ -phosphane, pentafluoridophosphorus			
PF ₆			PF ₆ ⁻ , hexafluoro-λ ⁵ -phosphanuide, hexafluoridophosphate(1-)	PF ₆ ⁻ , hexafluoro-λ ⁵ -phosphanuido, hexafluoridophosphato(1-)
PH	PH ^{2•} , phosphanylidene, hydridophosphorus(2•) >PH, phosphanediyl =PH, phosphanylidene	PH ^{•+} , phosphanyliumyl, hydridophosphorus(•1+) PH ²⁺ , phosphanebis(ylium), hydridophosphorus(2+)	PH ^{•-} , phosphanidyl, hydridophosphate(•1-) PH ²⁻ , phosphanediide, hydridophosphate(2-)	PH ²⁻ , phosphanediido, hydridophosphato(2-)
PH ₂	PH ₂ [•] , phosphanyl, dihydridophosphorus(•) -PH ₂ , phosphanyl	PH ₂ ⁺ , phosphanylium, dihydridophosphorus(1+)	PH ₂ ⁻ , phosphanide, dihydridophosphate(1-)	PH ₂ ⁻ , phosphanido, dihydridophosphato(1-)
PH ₃	PH ₃ , phosphorus trihydride [PH ₃], phosphane (parent hydride name), trihydridophosphorus	PH ₃ ^{•+} , phosphaniumyl, trihydridophosphorus(•1+) -PH ₃ ⁺ , phosphaniumyl	PH ₃ ^{•-} , phosphanuidyl, trihydridophosphate(•1-) ^e	PH ₃ , phosphane
PH ₄	-PH ₄ , λ ⁵ -phosphanyl	PH ₄ ⁺ , phosphanium, tetrahydridophosphorus(1+)	PH ₄ ⁻ , phosphanuide, tetrahydridophosphate(1-)	PH ₄ ⁻ , phosphanuido, tetrahydridophosphato(1-)
PH ₅	PH ₅ , phosphorus pentahydride [PH ₅], λ ⁵ -phosphane (parent hydride name), pentahydridophosphorus			
PN	P≡N, nitridophosphorus >P≡N, azanylidyne-λ ⁵ -phosphanediyl; phosphoronitridoyl			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
PO	PO [•] , oxophosphanyl, oxidophosphorus(•), phosphorus mon(o)oxide; phosphoryl >P(O)–, oxo-λ ⁵ -phosphanetriyl; phosphoryl =P(O)–, oxo-λ ⁵ -phosphanylidene; phosphorylidene ≡P(O), oxo-λ ⁵ -phosphanylidyne; phosphorylidyne	PO ⁺ , oxidophosphorus(1+) (not phosphoryl)	PO [–] , oxidophosphate(1–)	
PO ₂	–P(O) ₂ , dioxo-λ ⁵ -phosphanyl		PO ₂ [–] , dioxidophosphate(1–)	PO ₂ [–] , dioxidophosphato(1–)
PO ₃			PO ₃ [–] , trioxidophosphate(1–) PO ₃ ^{•2–} , trioxidophosphate(•2–) PO ₃ ^{3–} , trioxidophosphate(3–); phosphite (PO ₃ [–]) _n = {P(O) ₂ O} _n ^{n–} , catena-poly[(dioxidophosphate-μ- oxido)(1–)]; metaphosphate –P(O)(O [–]) ₂ , dioxidooxo-λ ⁵ -phosphanyl; phosphonato	PO ₃ [–] , trioxidophosphato(1–) PO ₃ ^{•2–} , trioxidophosphato(•2–) PO ₃ ^{3–} , trioxidophosphato(3–); phosphito

PO_4			$\text{PO}_4^{\bullet 2-}$, tetraoxidophosphate($\bullet 2-$) PO_4^{3-} , tetraoxidophosphate($3-$); phosphate	PO_4^{3-} , tetraoxidophosphato($3-$); phosphato
PO_5			$\text{PO}_5^{\bullet 2-} = \text{PO}_3(\text{OO})^{\bullet 2-}$, trioxidoperoxidophosphate($\bullet 2-$) $\text{PO}_5^{3-} = \text{PO}_3(\text{OO})^{3-}$, trioxidoperoxidophosphate($3-$); peroxyphosphato, phosphoroperoxoato	$\text{PO}_5^{3-} = \text{PO}_3(\text{OO})^{3-}$, trioxidoperoxidophosphato($3-$); peroxyphosphato, phosphoroperoxoato
PS	$\text{PS}^\bullet$, sulfidophosphorus($\bullet$); -PS, thiophosphoryl	PS^+ , sulfidophosphorus($1+$) (<i>not</i> thiophosphoryl)		
PS_4			PS_4^{3-} , tetrasulfidophosphate($3-$)	PS_4^{3-} , tetrasulfidophosphato($3-$)
P_2	P_2 , diphosphorus	P_2^+ , diphosphorus($1+$)	P_2^- , diphosphide($1-$) P_2^{2-} , diphosphide($2-$)	P_2 , diphosphorus P_2^- , diphosphido($1-$) P_2^{2-} , diphosphido($2-$)
P_2H			$\text{HP} = \text{P}^-$, diphosphenide PPH^{3-} , diphosphanetriide	$\text{HP} = \text{P}^-$, diphosphenido PPH^{3-} , diphosphanetriido
P_2H_2	$\text{HP} = \text{PH}$, diphosphene (parent hydride name) $\text{H}_2\text{PP}^{2\bullet}$, diphosphanylidene $= \text{PPH}_2$, diphosphanylidene $= \text{HPPH}^-$, diphosphane-1,2-diyl		HPPH^{2-} , diphosphane-1,2-diide H_2PP^{2-} , diphosphane-1,1-diide	$\text{HP} = \text{PH}$, diphosphene HPPH^{2-} , diphosphane-1,2-diido H_2PP^{2-} , diphosphane-1,1-diido
P_2H_3	$\text{H}_2\text{PPH}^\bullet$, diphosphanyl, trihydridodiphosphorus($P-P$)($\bullet$) - HPPH_2 , diphosphanyl		H_2PPH^- , diphosphanide	H_2PPH^- , diphosphanido
P_2H_4	H_2PPH_2 , diphosphane (parent hydride name)			H_2PPH_2 , diphosphane
P_2O_6			$\text{O}_3\text{PPO}_3^{3-}$, bis(trioxidophosphate)($P-P$)($4-$); hypodiphosphate	$\text{O}_3\text{PPO}_3^{3-}$, bis(trioxidophosphato)($P-P$)($4-$); hypodiphosphato
P_2O_7			$\text{O}_3\text{POPO}_3^{4-}$, μ -oxido- bis(trioxidophosphate)($4-$); diphosphate	$\text{O}_3\text{POPO}_3^{4-}$, μ -oxido- bis(trioxidophosphato)($4-$); diphosphato

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
P ₂ O ₈			O ₃ POOPO ₃ ⁴⁻ , μ-peroxido-1κO,2κO'- bis(trioxidophosphate)(4-); peroxydiphosphate	O ₃ POOPO ₃ ⁴⁻ , μ-peroxido-1κO,2κO'- bis(trioxidophosphate)(4-); peroxydiphosphate
P ₄	P ₄ , tetraphosphorus			P ₄ , tetraphosphorus
Pa	protactinium	protactinium	protactinide	protactinido
Pb	lead	lead (general) Pb ²⁺ , lead(2+) Pb ⁴⁺ , lead(4+)	plumbide	plumbido
PbH ₄	PbH ₄ , plumbane (parent hydride name), tetrahydrido-lead, lead tetrahydride			
Pb ₉			Pb ₉ ⁴⁻ , nonaplumbide(4-)	
Pd	palladium	palladium (general) Pd ²⁺ , palladium(2+) Pd ⁴⁺ , palladium(4+)	palladiide	palladiido
Pm	promethium	promethium	promethide	promethido
Po	polonium	polonium	polonide	polonido
PoH ₂ , see H ₂ Po				
Pr	praseodymium	praseodymium	praseodymide	praseodymido
Pt	platinum	platinum (general) Pt ²⁺ , platinum(2+) Pt ⁴⁺ , platinum(4+)	platinide	platinido

Pu	plutonium	plutonium	plutonide	plutonido
PuO ₂	PuO ₂ , plutonium dioxide	PuO ₂ ⁺ , dioxidoplutonium(1+) [not plutonyl(1+)] PuO ₂ ²⁺ , dioxidoplutonium(2+) [not plutonyl(2+)]		
Ra	radium	radium	radide	radio
Rb	rubidium	rubidium	rubidide	rubidido
Re	rhénium	rhénium	rhénide	rhénido
ReO ₄			ReO ₄ ⁻ , tetraoxidorhenate(1-) ReO ₄ ²⁻ , tetraoxidorhenate(2-)	ReO ₄ ⁻ , tetraoxidorhenato(1-) ReO ₄ ²⁻ , tetraoxidorhenato(2-)
Rf	rutherfordium	rutherfordium	rutherfordide	rutherfordido
Rg	roentgenium	roentgenium	roentgenide	roentgenido
Rh	rhodium	rhodium	rhodide	rhodido
Rn	radon	radon	radonide	radonido
Ru	ruthénium	ruthénium	ruthénide	ruthénido
S	sulfur (general) S, monosulfur =S, sulfanylidene; thioxo -S-, sulfanedyl	sulfur (general) S ⁺ , sulfur(1+)	sulfide (general) S ^{•-} , sulfamidyl, sulfide(•1-) S ²⁻ , sulfanedide, sulfide(2-); sulfide -S-, sulfido	sulfido (general) S ^{•-} , sulfamidyl, sulfido(•1-) S ²⁻ , sulfanedido, sulfido(2-)
SCN, see CNS				
SH, see HS				
SH ₂ , see H ₂ S				
SNC, see CNS				
SO	SO, sulfur mon(o)oxide [SO], oxidosulfur > SO, oxo-λ ⁴ -sulfanedyl; sulfinyl	SO ^{•+} , oxidosulfur(•1+) (not sulfinyl or thionyl)	SO ^{•-} , oxidosulfate(•1-)	[SO], oxidosulfur
SO ₂	SO ₂ , sulfur dioxide [SO ₂], dioxidosulfur >SO ₂ , dioxo-λ ⁶ -sulfanedyl; sulfonyl, sulfonyl		SO ₂ ^{•-} , dioxidosulfate(•1-) SO ₂ ²⁻ , dioxidosulfate(2-), sulfanediolate	[SO ₂], dioxidosulfur SO ₂ ²⁻ , dioxidosulfato(2-), sulfanediolato

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
SO ₃	SO ₃ , sulfur trioxide		SO ₃ ^{•-} , trioxidosulfate(•1-) SO ₃ ²⁻ , trioxidosulfate(2-); sulfite -S(O) ₂ (O ⁻), oxidodioxo-λ ⁶ -sulfanyl; sulfonato	SO ₃ ²⁻ , trioxidosulfato(2-); sulfito
SO ₄	-OS(O) ₂ O ⁻ , sulfonylbis(oxy)		SO ₄ ^{•-} , tetraoxidosulfate(•1-) SO ₄ ²⁻ , tetraoxidosulfate(2-); sulfate	SO ₄ ²⁻ , tetraoxidosulfato(2-); sulfato
SO ₅			SO ₅ ^{•-} = SO ₃ (OO) ^{•-} , trioxidoperoxidosulfate(•1-) SO ₅ ²⁻ = SO ₃ (OO) ²⁻ , trioxidoperoxidosulfate(2-); peroxysulfate, sulfuroperoxoate	SO ₅ ²⁻ = SO ₃ (OO) ²⁻ , trioxidoperoxidosulfato(2-); peroxysulfato, sulfuroperoxoato
S ₂	S ₂ , disulfur -SS-, disulfanediyl >S=S, sulfanylidene-λ ⁴ -sulfanediyl; sulfinothioyl	S ₂ ^{•+} , disulfur(•1+)	S ₂ ^{•-} , disulfanidylyl, disulfide(•1-) S ₂ ²⁻ , disulfide(2-), disulfanediide -SS-, disulfanidylyl	S ₂ ²⁻ , disulfido(2-), disulfanediido
S ₂ O	>S(=O)(=S), oxosulfanylidene-λ ⁶ -sulfanediyl; sulfonothioyl			

S_2O_2			$S_2O_2^{2-} = OSSO_2^{2-}$, disulfanediolate, bis(oxidosulfate)($S-S$)(2-) $S_2O_2^{2-} = SOOS^{2-}$, dioxidanedithiolate, peroxybis(sulfanide), bis(sulfidoxygenate)($O-O$)(2-) $S_2O_2^{2-} = SO_2S^{2-}$, dioxido-1 κ^2O -disulfate($S-S$)(2-), dioxidosulfidosulfate(2-); thiosulfite, sulfurothioite	$S_2O_2^{2-} = OSSO_2^{2-}$, disulfanediolate, bis(oxidosulfate)($S-S$)(2-) $S_2O_2^{2-} = SOOS^{2-}$, dioxidanedithiolate, peroxybis(sulfanide), bis(sulfidoxygenate)($O-O$)(2-) $S_2O_2^{2-} = SO_2S^{2-}$, dioxido-1 κ^2O -disulfate($S-S$)(2-), dioxidosulfidosulfate(2-); thiosulfite, sulfurothioite
S_2O_3			$S_2O_3^{\bullet-} = SO_3S^{\bullet-}$, trioxido-1 κ^3O -disulfate($S-S$)(•1-), trioxidosulfidosulfate(•1-) $S_2O_3^{2-} = SO_3S^{2-}$, trioxido-1 κ^3O -disulfate($S-S$)(2-), trioxidosulfidosulfate(2-); thiosulfate, sulfurothioate	$S_2O_3^{2-} = SO_3S^{2-}$, trioxido-1 κ^3O -disulfate($S-S$)(2-), trioxidosulfidosulfate(2-); thiosulfate, sulfurothioate
S_2O_4			$S_2O_4^{2-} = O_2SSO_2^{2-}$, bis(dioxidosulfate)($S-S$)(2-); dithionite	$S_2O_4^{2-} = O_2SSO_2^{2-}$, bis(dioxidosulfate)($S-S$)(2-); dithionite
S_2O_5			$S_2O_5^{2-} = O_3SSO_2^{2-}$, pentaoxido- 1 $\kappa^3O, 2\kappa^2O$ -disulfate($S-S$)(2-) $S_2O_5^{2-} = O_2SOSO_2^{2-}$, μ -oxido-bis(dioxidosulfate)(2-) $S_2O_6^{2-} = O_3SSO_3^{2-}$, bis(trioxidosulfate)($S-S$)(2-); dithionate	$S_2O_5^{2-} = O_3SSO_2^{2-}$, pentaoxido- 1 $\kappa^3O, 2\kappa^2O$ -disulfate($S-S$)(2-) $S_2O_5^{2-} = O_2SOSO_2^{2-}$, μ -oxido-bis(dioxidosulfate)(2-) $S_2O_6^{2-} = O_3SSO_3^{2-}$, bis(trioxidosulfate)($S-S$)(2-); dithionate
S_2O_6				
S_2O_7			$S_2O_7^{2-} = O_3SOSO_3^{2-}$, μ -oxido-bis(trioxidosulfate)(2-); disulfate	$S_2O_7^{2-} = O_3SOSO_3^{2-}$, μ -oxido-bis(trioxidosulfate)(2-); disulfate

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
S_2O_8			$S_2O_8^{2-} = O_3SOOSO_3^{2-}$, μ -peroxido-1κO,2κO'-bis(trioxidosulfate)(2-); peroxydisulfate	$S_2O_8^{2-} = O_3SOOSO_3^{2-}$, μ -peroxido-1κO,2κO'-bis(trioxidosulfate)(2-); peroxydisulfate
S_3	S_3 , trisulfur -SSS-, trisulfaneyl >S(=S) ₂ , bis(sulfanylidene)-λ ⁶ -sulfaneyl; sulfonodithioyl, dithiosulfonyl	S_3^{2+} , trisulfur(2+)	$S_3^{\bullet-}$, trisulfide(•1-) SSS ^{•-} , trisulfaneyl S_3^{2-} , trisulfide(2-) SSS ²⁻ , trisulfaneylide	$S_3^{\bullet-}$, trisulfido(•1-) SSS ^{•-} , trisulfaneyl S_3^{2-} , trisulfido(2-) SSS ²⁻ , trisulfaneydiido
S_4	S_4 , tetrasulfur -SSSS-, tetrasulfaneyl	S_4^{2+} , tetrasulfur(2+)	S_4^{2-} , tetrasulfide(2-) SSSS ²⁻ , tetrasulfaneydiide	S_4^{2-} , tetrasulfido(2-) SSSS ²⁻ , tetrasulfaneydiido
S_4O_6			$S_4O_6^{2-} = O_3SSSO_3^{2-}$, disulfaneydisulfonate, bis[(trioxidosulfato)sulfate]⊖ (S-S)(2-); tetrathionate $S_4O_6^{\bullet 3-} = O_3SSSO_3^{\bullet 3-}$, bis[(trioxidosulfato)sulfate]⊖ (S-S)(•3-)	$S_4O_6^{2-} = O_3SSSO_3^{2-}$, disulfaneydisulfonate, bis[(trioxidosulfato)sulfate]⊖ (S-S)(2-); tetrathionate $S_4O_6^{\bullet 3-} = O_3SSSO_3^{\bullet 3-}$, bis[(trioxidosulfato)sulfate]⊖ (S-S)(•3-)
S_5	S_5 , pentasulfur		S_5^{2-} , pentasulfide(2-) SSSSS ²⁻ , pentasulfaneydiide	S_5^{2-} , pentasulfido(2-) SSSSS ²⁻ , pentasulfaneydiido
S_8	S_8 , octasulfur	S_8^{2+} , octasulfur(2+)	S_8^{2-} , octasulfide(2-) S[S] ₆ S ²⁻ , octasulfaneydiide	S_8 , octasulfur S_8^{2-} , octasulfido(2-) S[S] ₆ S ²⁻ , octasulfaneydiido

Sb	antimony >Sb ⁻ , stibanetriyl	antimony	antimonide (general) Sb ³⁻ , antimonide(3-), stibanetriide; antimonide	antimonido (general) Sb ³⁻ , antimonido, stibanetriido
SbH	SbH [•] , stibanylidene, hydridoantimony(2•) >SbH, stibanediyl =SbH, stibanylidene	SbH ²⁺ , stibanebis(ylum), hydridoantimony(2+)	SbH ²⁻ , stibanediide, hydridoantimonate(2-)	SbH ²⁻ , stibanediido, hydridoantimonato(2-)
SbH ₂	SbH ₂ [•] , stibanyl, dihydridoantimony(•) -SbH ₂ , stibanyl	SbH ₂ ⁺ , stibanylium, dihydridoantimony(1+)	SbH ₂ ⁻ , stibanide, dihydridoantimonate(1-)	SbH ₂ ⁻ , stibanido, dihydridoantimonato(1-)
SbH ₃	SbH ₃ , antimony trihydride [SbH ₃], stibane (parent hydride name), trihydridoantimony -SbH ₄ , λ ⁵ -stibanyl	SbH ₃ ^{•+} , stibaniumyl, trihydridoantimony(•1+) -SbH ₃ ⁺ , stibaniumyl	SbH ₃ ^{•-} , stibanuidyl, trihydridoantimonate(•1-) ^e	SbH ₃ , stibane
SbH ₄		SbH ₄ ⁺ , stibanium, tetrahydridoantimony(1+)		
SbH ₅	SbH ₅ , antimony pentahydride [SbH ₅], λ ⁵ -stibane (parent hydride name), pentahydridoantimony			
Sc	scandium	scandium	scandide	scandido
Se	Se (general) Se, monoselenium >Se, selanediyl =Se, selanylidene; selenoxo	selenium	selenide (general) Se ^{•-} , selanidyl, selenide(•1-) Se ²⁻ , selanediide, selenide(2-); selenide	selenido (general) Se ^{•-} , selanidyl, selenido(•1-) Se ²⁻ , selanediido, selenido(2-)
SeCN, see CNSe				
SeH, see HSe				
SeH ₂ , see H ₂ Se				
SeO	SeO, selenium mon(o)oxide [SeO], oxidoselenium > SeO, seleninyl			[SeO], oxidoselenium

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
SeO ₂	SeO ₂ , selenium dioxide [SeO ₂], dioxidoselenium >SeO ₂ , selenonyl		SeO ₂ ²⁻ , dioxidoselenate(2-)	[SeO ₂], dioxidoselenium SeO ₂ ²⁻ , dioxidoselenato(2-)
SeO ₃	SeO ₃ , selenium trioxide		SeO ₃ ^{•-} , trioxidoselenate(•1-) SeO ₃ ²⁻ , trioxidoselenate(2-); selenite	SeO ₃ ²⁻ , trioxidoselenato(2-); selenito
SeO ₄			SeO ₄ ²⁻ , tetraoxidoselenate(2-); selenate	SeO ₄ ²⁻ , tetraoxidoselenato(2-); selenato
Sg	seaborgium	seaborgium	seaborgide	seaborgido
Si	silicon >Si<, silanetetrayl =Si=, silanediylidene	silicon (general) Si ⁺ , silicon(•1+) Si ⁴⁺ , silicon(4+)	silicide (general) Si ⁻ , silicide(•1-) Si ⁴⁻ , silicide(4-); silicide	silicido (general) Si ⁻ , silicido(•1-) Si ⁴⁻ , silicido(4-); silicido
SiC	SiC, silicon carbide [SiC], carbidosilicon	SiC ⁺ , carbidosilicon(1+)		
SiH		SiH ⁺ , silanyliumdiyl, hydridosilicon(1+)	SiH ⁻ , silanediyl, hydridosilicate(1-)	
SiH ₂	SiH ₂ ^{2•} , silylidene, dihydridosilicon(2•) >SiH ₂ , silanediyl =SiH ₂ , silylidene			
SiH ₃	SiH ₃ [•] , silyl, trihydridosilicon(•) -SiH ₃ , silyl	SiH ₃ ⁺ , silylium, trihydridosilicon(1+)	SiH ₃ ⁻ , silanide, trihydridosilicate(1-)	SiH ₃ ⁻ , silanido

SiH ₄	SiH ₄ , silicon tetrahydride [SiH ₄], silane (parent hydride name), tetrahydrosilicon			
SiO	SiO, oxidosilicon, silicon mon(o)oxide	SiO ⁺ , oxidosilicon(1+)		
SiO ₂	SiO ₂ , silicon dioxide			
SiO ₃			SiO ₃ ^{•-} , trioxidosilicate(•1-) (SiO ₃ ²⁻) _n = tSi(O) ₂ O ₃ ²ⁿ⁻ , <i>catena</i> -poly[(dioxidosilicate- μ-oxido)(1-)]; metasilicate	SiO ₃ ^{•-} , trioxidosilicato(•1-)
SiO ₄			SiO ₄ ⁴⁻ , tetraoxidosilicate(4-); silicate	SiO ₄ ⁴⁻ , tetraoxidosilicato(4-); silicato
Si ₂	Si ₂ , disilicon	Si ₂ ⁺ , disilicon(1+)	Si ₂ ⁻ , disilicide(1-)	
Si ₂ H ₄	>SiHSiH ₃ , disilane-1,1-diyl -SiH ₂ SiH ₂ ⁻ , disilane-1,2-diyl = SiHSiH ₃ , disilanylidene			
Si ₂ H ₅	Si ₂ H ₅ [•] , disilanyl, pentahydridosilicon(Si—Si)(•) -Si ₂ H ₅ , disilanyl	Si ₂ H ₅ ⁺ , disilanylium	Si ₂ H ₅ ⁻ , disilanide	Si ₂ H ₅ ⁻ , disilanido
Si ₂ H ₆	Si ₂ H ₆ , disilane (parent hydride name)			Si ₂ H ₆ , disilane
Si ₂ O ₇			Si ₂ O ₇ ⁶⁻ , μ-oxido-bis(trioxidosilicate)(6-); disilicate	Si ₂ O ₇ ⁶⁻ , μ-oxido-bis(trioxidosilicato)(6-); disilicato
Si ₄			Si ₄ ⁴⁻ , tetrasilicide(4-)	
Sm	samarium	samarium	samaride	samarido
Sn	tin	tin (general) Sn ²⁺ , tin(2+) Sn ⁴⁺ , tin(4+)	stannide	stannido
SnCl ₃			SnCl ₃ ⁻ , trichloridostannate(1-)	SnCl ₃ ⁻ , trichloridostannato(1-)
SnH ₄	SnH ₄ , tin tetrahydride [SnH ₄], stannane (parent hydride name), tetrahydridotin			

Table IX Continued

Name				
Formula for uncharged atom or group	Uncharged atoms or molecules (including zwitterions and radicals) or substituent groups ^a	Cations (including cation radicals) or cationic substituent groups ^a	Anions (including anion radicals) or anionic substituent groups ^b	Ligands ^c
Sn ₅			Sn ₅ ²⁻ , pentastannide(2-)	Sn ₅ ²⁻ , pentastannido(2-)
Sr	strontium	strontium	strontide	strontido
T, see H				
T ₂ , see H ₂				
T ₂ O, see H ₂ O				
Ta	tantalum	tantalum	tantalide	tantalido
Tb	terbium	terbium	terbide	terbido
Tc	technetium	technetium	technetide	technetido
TcO ₄			TcO ₄ ⁻ , tetraoxido technetate(1-) TcO ₄ ²⁻ , tetraoxido technetate(2-)	TcO ₄ ⁻ , tetraoxido technetato(1-) TcO ₄ ²⁻ , tetraoxido technetato(2-)
Te	tellurium >Te, tellanediyl =Te, tellanylidene; telluroxo	tellurium	telluride (general) Te ^{•-} , tellanidylyl, telluride(•1-) Te ²⁻ , tellanediide, telluride(2-); telluride	tellurido (general) Te ^{•-} , tellanidylyl, tellurido(•1-) Te ²⁻ , tellanediido, tellurido(2-)
TeH, see HTe				
TeH ₂ , see H ₂ Te				
TeO ₃			TeO ₃ ^{•-} , trioxido tellurate(•1-) TeO ₃ ²⁻ , trioxido tellurate(2-)	TeO ₃ ^{•-} , trioxido tellurato(•1-) TeO ₃ ²⁻ , trioxido tellurato(2-)
TeO ₄			TeO ₄ ²⁻ , tetraoxido tellurate(2-); tellurate	TeO ₄ ²⁻ , tetraoxido tellurato(2-); tellurato
TeO ₆			TeO ₆ ⁶⁻ , hexaoxido tellurate(6-); orthotellurate	TeO ₆ ⁶⁻ , hexaoxido tellurato(6-); orthotellurato
Th	thorium	thorium	thoride	thorido

Ti	titanium	titanium	titanide	titanido
TiO	TiO, titanium(II) oxide	TiO ²⁺ , oxidotitanium(2+)		
Tl	thallium	thallium	thallide	thallido
TlH ₂	-TlH ₂ , thallanyl			
TlH ₃	TlH ₃ , thallium trihydride [TlH ₃], thallane (parent hydride name), trihydridothallium			
Tm	thulium	thulium	thulide	thulido
U	uranium	uranium	uranide	uranido
UO ₂	UO ₂ , uranium dioxide	UO ₂ ⁺ , dioxidouranium(1+) [<i>not</i> uranyl(1+)] UO ₂ ²⁺ , dioxidouranium(2+) [<i>not</i> uranyl(2+)]		
V	vanadium	vanadium	vanadide	vanadido
VO	VO, vanadium(II) oxide, vanadium mon(o)oxide	VO ²⁺ , oxidovanadium(2+) (<i>not</i> vanadyl)		
VO ₂	VO ₂ , vanadium(IV) oxide, vanadium dioxide	VO ₂ ⁺ , dioxidovanadium(1+)		
W	tungsten	tungsten	tungstide	tungstido
Xe	xenon	xenon	xenonide	xenonido
Y	yttrium	yttrium	yttride	yttrido
Yb	ytterbium	ytterbium	ytterbide	ytterbido
Zn	zinc	zinc	zincide	zincido
Zr	zirconium	zirconium	zirconide	zirconido
ZrO	ZrO, zirconium(II) oxide	ZrO ²⁺ , oxidozirconium(2+)		

^a Where an element symbol occurs in the first column, the unmodified element name is listed in the second and third columns. The unmodified name is generally used when the element appears as an electropositive constituent in the construction of a stoichiometric name (Sections IR-5.2 and IR-5.4). Names of homoatomic cations consisting of the element are also constructed using the element name, adding multiplicative prefixes and charge numbers as applicable (Sections IR-5.3.2.1 to IR-5.3.2.3). In selected cases, examples are given in the Table of specific cation names, such as gold(1+), gold(3+); mercury(2+), dimercury(2+). In such cases, the unmodified element name appears with the qualifier '(general)'.

- ^b Where an element symbol occurs in the first column, the fourth column gives the element name appropriately modified with the ending 'ide' (argentide, americide, ferride, etc.). The 'ide' form of the element name is generally used when the element appears as an electronegative constituent in the construction of a stoichiometric name (Sections IR-5.2 and IR-5.4). Names of homoatomic anions consisting of the element in question are also constructed using this modified form, adding multiplicative prefixes and charge numbers as applicable (Sections IR-5.3.3.1 to IR-5.3.3.3). Examples are given in the Table of names of some specific anions, e.g. arsenide(3-), chloride(1-), oxide(2-), dioxide(2-). In certain cases, a particular anion has the 'ide' form itself as an accepted short name, e.g. arsenide, chloride, oxide. If specific anions are named, the 'ide' form of the element name with no further modification is given as the first entry in the fourth column, with the qualifier '(general)'.
- ^c Ligand names must be placed within enclosing marks whenever necessary to avoid ambiguity, cf. Section IR-9.2.2.3. Some ligand names must always be enclosed. For example, if 'dioxido' is cited as it must be enclosed so as to distinguish it from two 'oxido' ligands; if combined with a multiplicative prefix it must be enclosed because it starts with a multiplicative prefix itself. A ligand name such as 'nitridocarbonato' must always be enclosed to avoid interpreting it as two separate ligand names, 'nitrido' and 'carbonato'. In this table, however, these enclosing marks are omitted for the sake of clarity. Note that the ligand names given here with a charge number can generally also be used without if it is not desired to make any implication regarding the charge of the ligand. For example, the ligand name '[dioxido(•1-)]' may be used if one wishes explicitly to consider the ligand to be the species dioxido(•1-), whereas the ligand name '(dioxido)' can be used if no such implications are desirable.
- ^d The ending 'ide' in 'actinide' and 'lanthanide' indicates a negative ion. Therefore, 'actinoid' should be used as the collective name for the elements Ac, Th, Pa, U, Np, Pu, Am, Cm, Bk, Cf, Es, Fm, Md, No, Lr, and 'lanthanoid' as a collective name for the elements La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu (cf. Section IR-3.5).
- ^e The radical names in the present recommendations sometimes differ from those given in "Names for Inorganic Radicals", W.H. Koppenol, *Pure Appl. Chem.*, **72**, 437-446 (2000). Firstly, the exceptional status of anion radicals consisting of hydrogen and only one other element has been lifted. For example, the coordination-type additive name of $\text{BH}_3^{\bullet-}$ is 'trihydroborate(•1-)' (not 'boride'). Secondly, concatenation of ligand names, such as in 'hydridodioxido' (meaning the ligand 'dioxidanido'), which is otherwise never used in additive nomenclature, is not recommended here. Thirdly, additive names of dinuclear compounds are based here on selecting the most centrally placed atoms in the molecule as central atoms (see the general principles described in Section IR-7.1.2), e.g. $\text{NCSSCN}^{\bullet-}$ is named here 'bis[cyanidosulfur](S-S)(•1-)' rather than 'bis(nitridosulfidocarbonato)(S-S)(•1-)'.
- ^f Due to the strict adherence in these recommendations to the element sequence in Table VI, the order of oxygen and the elements chlorine, bromine and iodine, respectively, has been reversed relative to traditional names. This applies to binary stoichiometric names such as dioxigen chloride (cf. Section IR-5.2) and to additive names for the hypohalites, where the rules for selecting central atoms (Section IR-7.1.2) dictate the selection of oxygen rather than the halide. However, because of the additive names for the last three members of the series OX^- , XO_2^- , XO_3^- , XO_4^- ($\text{X} = \text{Cl}, \text{Br}, \text{I}$), namely dioxidohalogenate(1-), trioxidohalogenate(1-) and tetraoxidohalogenate(1-) (the halogen is chosen as the central atom because it has the central position in the structure), the additive names oxidochlorate(1-), oxidobromate(1-) and oxidoiodate(1-) are acceptable alternatives to the systematic 'oxygenate' names. Similar remarks apply to HOCl , $\text{HOCl}^{\bullet-}$, etc.

Table X Anion names, 'a' terms used in substitutive nomenclature and 'y' terms used in chains and rings nomenclature

<i>Element name</i>	<i>Anion name^a</i>	<i>'a' term</i>	<i>'y' term</i>
actinium	actinate	actina	actiny
aluminium	aluminate	alumina	aluminy
americium	americite	america	americy
antimony	antimonate	stiba ^b	stiby ^b
argon	argonate	argona	argony
arsenic	arsenate	arsa	arsy
astatine	astatate	astata	astaty
barium	barate	bara	bary
berkelium	berkelate	berkela	berkely
beryllium	beryllate	berylla	beryllly
bismuth	bismuthate	bisma	bismy
bohrium	bohrate	bohra	bohry
boron	borate	bora	bory
bromine	bromate	broma	bromy
cadmium	cadmate	cadma	cadmy
caesium	caesate	caesa	caesy
calcium	calcate	calca	calcy
californium	californate	california	californy
carbon	carbonate	carba	carby
cerium	cerate	cera	cery
chlorine	chlorate	chlora	chlory
chromium	chromate	chroma	chromy
cobalt	cobaltate	cobalta	cobalty
copper	cuprate ^c	cupra ^c	cupry ^c
curium	curate	cura	cury
darmstadtium	darmstadtate	darmstadta	darmstadty
deuterium	deuterate	deutera	deutery
dubnium	dubnate	dubna	dubny
dysprosium	dysprosate	dysprosa	dysprosy
einsteinium	einsteinate	einsteina	einsteinly
erbium	erbate	erba	erby
europium	europate	europa	europy
fermium	fermate	ferma	fermy
fluorine	fluorate	fluora	fluory
francium	francate	franca	francy
gadolinium	gadolate	gadolina	gadolinly
gallium	gallate	galla	gally
germanium	germanate	germa	germy
gold	aurate ^d	aura ^d	aury ^d
hafnium	hafnate	hafna	hafny
hassium	hassate	hassa	hassy
helium	helate	hela	hely
holmium	holmate	holma	holmy
hydrogen	hydrogenate	—	hydrony
indium	indate	inda	indy
iodine	iodate	ioda	iody
iridium	iridate	irida	iridy
iron	ferrate ^c	ferra ^c	ferry ^c
krypton	kryptonate	kryptona	kryptony
lanthanum	lanthanate	lanthana	lanthany
lawrencium	lawrencate	lawrenca	lawrenly

Table X *Continued*

<i>Element name</i>	<i>Anion name^a</i>	<i>'a' term</i>	<i>'y' term</i>
lead	plumbate ^f	plumba ^f	plumby ^f
lithium	lithate	litha	lithy
lutetium	lutetate	luteta	lutety
magnesium	magnesate	magnesa	magnesy
manganese	manganate	mangana	mangany
meitnerium	meitnerate	meitnera	meitnery
mendelevium	mendelevate	mendeleva	mendelevy
mercury	mercurate	mercura	mercury
molybdenum	molybdate	molybda	molybdy
neodymium	neodymate	neodyma	neodymy
neon	neonate	neona	neony
neptunium	neptunate	neptuna	neptuny
nickel	nickelate	nickela	nickely
niobium	niobate	nioba	nioby
nitrogen	nitrate	aza ^g	azy ^g
nobelium	nobelate	nobela	nobely
osmium	osmate	osma	osmy
oxygen	oxygenate	oxa	oxy
palladium	palladate	pallada	pallady
phosphorus	phosphate	phospha	phosphy
platinum	platinate	platina	platiny
plutonium	plutonate	plutona	plutony
polonium	polonate	polona	polony
potassium	potassate	potassa	potassy
praseodymium	praseodymate	praseodyma	praseodymy
promethium	promethate	prometha	promethy
protactinium	protactinate	protactina	protactiny
protium	protate	prota	proty
radium	radate	rada	rady
radon	radonate	radona	radony
rhenium	rhenate	rhena	rheny
rhodium	rhodate	rhoda	rhody
roentgenium	roentgenate	roentgena	roentgeny
rubidium	rubidate	rubida	rubidy
ruthenium	ruthenate	ruthena	rutheny
rutherfordium	rutherfordate	rutherforda	rutherfordy
samarium	samarate	samara	samary
scandium	scandate	scanda	scandy
seaborgium	seaborgate	seaborga	seaborgy
selenium	selenate	selena	seleny
silicon	silicate	sila	sily
silver	argentate ^h	argenta ^h	argenty ^h
sodium	sodate	soda	sody
strontium	strontate	stronta	stronty
sulfur	sulfate	thia ⁱ	sulfy
tantalum	tantalate	tantala	tantaly
technetium	technetate	techneta	technety
tellurium	tellurate	tellura	tellury
terbium	terbate	terba	terby
thallium	thallate	thalla	thally
thorium	thorate	thora	thory
thulium	thulate	thula	thuly

Table X *Continued*

<i>Element name</i>	<i>Anion name^a</i>	<i>'a' term</i>	<i>'y' term</i>
tin	stannate ^j	stanna ^j	stanny ^j
titanium	titanate	titana	titany
tritium	tritrate	trita	trity
tungsten	tungstate	tungsta	tungsty ^k
uranium	uranate	urana	urany
vanadium	vanadate	vanada	vanady
xenon	xenonate	xenona	xenony
ytterbium	ytterbate	ytterba	ytterby
yttrium	yttrate	yttra	yttry
zinc	zincate	zinca	zincy
zirconium	zirconate	zircona	zircony

^a Modified element name used in additive names for heteroatomic anions containing the element as the central atom.

^b From the name stibium.

^c From the name cuprum.

^d From the name aurum.

^e From the name ferrum.

^f From the name plumbum.

^g From the name azote.

^h From the name argentum.

ⁱ From the name theion.

^j From the name stannum.

^k 'Wolframy' was used in "Nomenclature of Inorganic Chains and Ring Compounds", E.O. Fluck and R.S. Laitinen, *Pure Appl. Chem.*, **69**, 1659–1692 (1997)" and in Chapter II-5 of *Nomenclature of Inorganic Chemistry II, IUPAC Recommendations 2000*, eds. J.A. McCleverty and N.G. Connelly, Royal Society of Chemistry, 2001.

Subject Index

Element names, parent hydride names and systematic names derived using any of the nomenclature systems described in this book are, with very few exceptions, not included explicitly in this index. If a name or term is referred to in several places in the book, only the more informative references may be indexed.

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